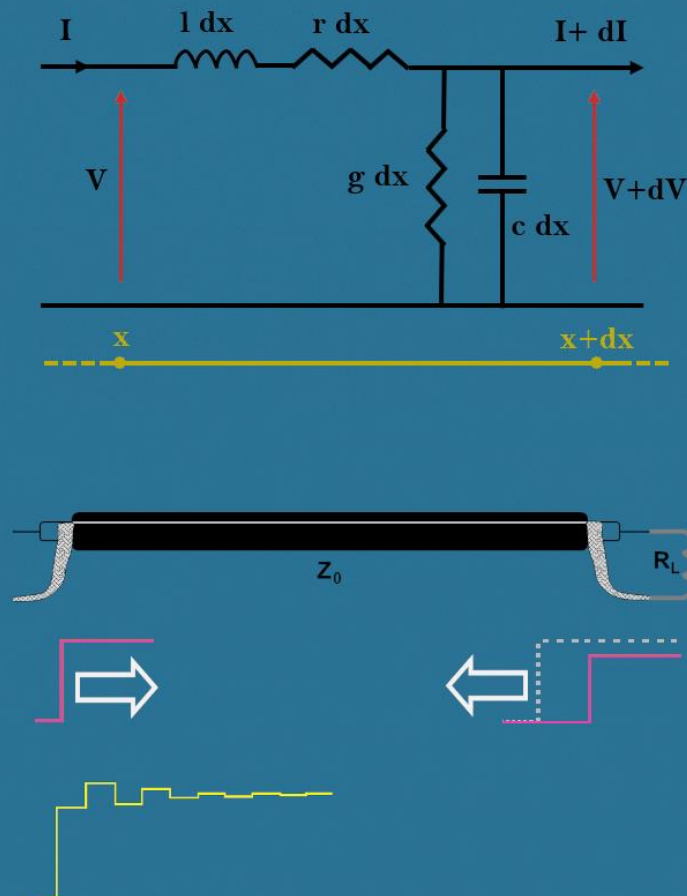


Discovering Invisible Reflections

A Journey into Time Domain Reflectometry



Leonardo Chieco

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Time Domain Reflectometry (TDR) is one of the most powerful and fascinating techniques for “seeing” what normally remains hidden inside a cable or transmission line. With a simple pulse or step signal, it is possible to reveal discontinuities, impedance mismatches, and variations that affect signal propagation.

In this tutorial, we will embark on a journey starting from the theoretical foundations of transmission lines and the reflection coefficient, leading up to the bounce diagram and practical simulations using LTspice. Concrete examples will be analyzed, from cases with open or short-circuited loads to more complex configurations with multiple lines, showing how to interpret the results and how to use them to diagnose real-world issues.

The goal is to guide the reader toward a deep understanding of TDR by connecting theory, simulation, and experimental practice, enabling its use as a tool for analysis and design in the fields of electronics, telecommunications, and system diagnostics.

Enjoy the reading!

The Author

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1. Introduzione

Time Domain Reflectometry, known by the acronym TDR, is a measurement technique that originated in the 1960s and is still widely used today in many fields of electrical engineering, electronics, and telecommunications. The underlying idea is conceptually simple yet extremely powerful: a very fast voltage or current edge is sent along a transmission line, and the way the signal is reflected by any discontinuities it encounters is observed. From the return time of the reflection, it is possible to determine the distance to the anomaly, while the amplitude and shape of the reflected wave reveal the nature of the discontinuity.

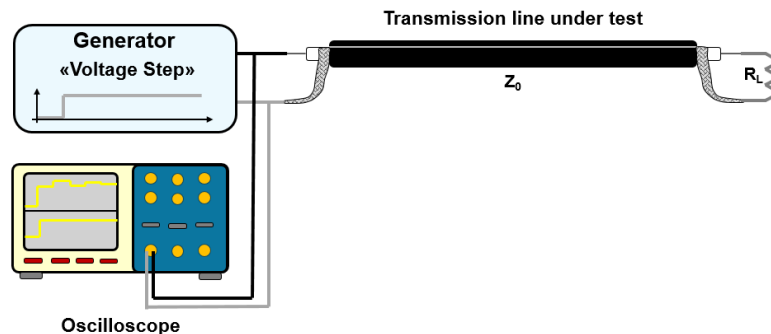


Figure 1: Principle of TDR operation

TDR can therefore be compared to a small “electrical radar”: instead of electromagnetic waves radiated into space, a guided pulse is used along a cable or a printed circuit board trace, and instead of locating airplanes or obstacles, it identifies opens, short circuits, impedance changes, faulty connections, or imperfectly matched junctions. This approach has made TDR a fundamental tool both in the laboratory and in the field, for applications ranging from coaxial cable diagnostics to high-speed PCB quality control.

One of the main reasons why time domain reflectometry is so important is the increasing need for signal integrity in modern systems. High-speed digital communications, RF electronics, differential buses on motherboards and memory interfaces all require transmission lines with extremely well-controlled impedance. Even a small mismatch can generate reflections that degrade the waveform, causing communication errors, jitter, and reduced signal margin.

Moreover, TDR is not only useful during development and characterization, but also for maintenance and diagnostics. An underground cable with a break can be inspected without digging it up or replacing it entirely: a simple TDR measurement is enough to determine how far from the injection point the anomaly is located. Similarly, in automotive wiring harnesses, aircraft electrical systems, or coaxial cables in telecommunications installations, TDR drastically reduces troubleshooting time.

There is also an economic and environmental aspect: accurately locating a fault means repairing only what is necessary, avoiding large-scale and costly replacements.

The key concept behind TDR is reflection due to impedance mismatch. An ideal transmission line, if perfectly matched to the load, generates no reflections: all the power is absorbed by the load. But if the load has an impedance different from the characteristic impedance of the line, part of the signal is reflected back toward the source. Measuring this reflection allows calculation of the so-called reflection coefficient, which indicates the magnitude and direction of the phenomenon.

Spatial information, i.e., the position of the discontinuity, is obtained by correlating the time-of-flight of the reflected signal with the propagation velocity of the electromagnetic wave in the medium. Since the velocity depends on the dielectric constant of the insulating material, TDR can also provide information about the physical properties of the transmission medium. Not surprisingly, beyond electronic applications, the technique is widely used in geotechnics and agriculture to measure soil moisture, since water alters the dielectric constant of the ground.

The versatility of time domain reflectometry is reflected in its wide range of applications:

1. Telecommunications and networks

- Coaxial cables: TDR is used to locate opens, shorts, or faulty junctions along RF cables.
- Copper data transmission cables (Ethernet, telephony): termination quality and the presence of crosstalk or discontinuities can be easily diagnosed.

2. High-speed digital electronics

- PCBs (Printed Circuit Boards): characterization of transmission lines on multilayer boards requires verification of characteristic impedance. TDR allows measurement of variations due to trace width differences, dielectric changes, or manufacturing defects.
- High-speed interfaces (DDR, PCIe, USB, HDMI): even minimal reflections can compromise proper operation, making TDR an essential tool for debugging and validation.

3. Aerospace and automotive

- Complex wiring: in avionics and automotive systems, identifying an electrical fault requires fast, non-invasive techniques. TDR allows fault localization without disassembling the entire harness.

4. Energy and electrical systems

- Power cables: a TDR measurement can identify points where insulation is degraded, locating the problem before it turns into a catastrophic failure.

5. Scientific and industrial applications

- Geotechnics and agriculture: TDR probes are inserted into the ground to measure water content. The variation in signal propagation velocity is correlated with moisture level.
- Quality control: during cable or PCB manufacturing, TDR is used to ensure that impedance remains within specified limits along the entire product length.

Types of TDR

There are two types of TDR, which differ in the type of signal used to excite the transmission line:

- Ideal pulse source (e.g., Dirac generator or very short rectangular pulse)
- Step source, which is the one normally assumed in TDR theory.

Characteristic	Pulse	Step
Signal shape	Short pulse with rapid rising and falling edges.	A rising edge bringing voltage from 0 to a constant value.
Available energy	Limited: all concentrated in a single packet. After the transition, nothing remains.	Practically infinite in time; the source continues feeding the line.
Response with matched load	The pulse reaches the load, is absorbed, and the voltage returns to zero.	The step reaches the load and remains stable at the value determined by the R_S - Z_L divider. The voltage at the load remains constant.
Reflection visibility	Reflections appear as secondary pulses following the main one, with amplitude proportional to the reflection coefficients.	Reflections appear as additional jumps or steps that change the voltage level at the load over time.
Practical interpretation	Requires instruments with high sensitivity and an excellent signal-to-noise ratio, because the reflected packets can have very small amplitudes.	Very intuitive: the voltage rises to a level, then any reflections cause it to "step" to the final value.
Typical use	Special techniques for rapid characterization, or when you want to directly measure the system's pulse response (similar to a frequency analysis).	It is the standard in classical TDR because it clearly shows mismatches and electrical lengths in the form of level variations.
Advantages	High temporal resolution, suitable for revealing fine details (equivalent to stimulating with a very wide bandwidth).	Easy to interpret, it does not require sophisticated instruments; reflections are visually added.
Disadvantages	Difficult to generate and measure with good amplitude; interpretation is more complex because it requires analysing pulse sequences.	Temporal resolution is limited by the rising edge of the step; it does not directly provide the frequency response.

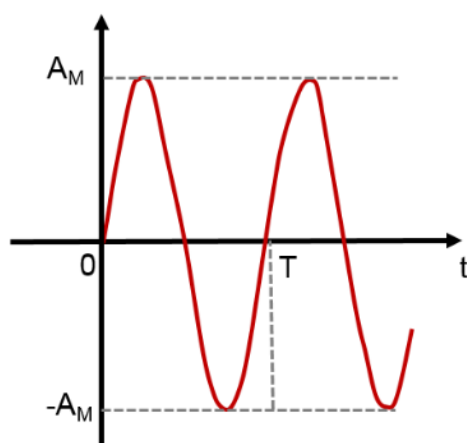
2. Review of Transmission Line Theory

In this chapter, fundamental mathematical tools for rigorous understanding of TDR will be introduced, such as sinusoidal steady-state, phasors, complex numbers, and derivatives. These are basic concepts of electrical engineering and mathematical analysis, essential for accurately describing wave behavior on a transmission line. It is worth noting that a deep mastery of all the theory is not strictly necessary: the most important concepts will always be summarized and explained in an intuitive way.

Those who wish to avoid the more mathematical details can skip this section without compromising understanding of the rest of the tutorial.

The study of electrical circuits often involves analyzing quantities that vary over time in arbitrary ways: rectangular edges, pulse signals, complex periodic functions. However, there is a special case so important that it deserves its own theory, known as sinusoidal steady-state.

We speak of sinusoidal steady-state when all electrical quantities (voltages and currents) vary sinusoidally in time and share the same frequency. They may differ in amplitude and phase, but they share a common frequency “ f ”. This occurs, for example, when a linear circuit is excited by a sinusoidal source: after the initial transient, only the forced response at fixed frequency remains.



$$a(t) = A_M \cos(\omega t + \alpha)$$

A_M = Maximum amplitude

α = initial phase ($-\pi < \alpha \leq \pi$ radians)

ω = angular frequency = $2\pi f$

T = period (seconds) = $\frac{2\pi}{\omega}$

f = frequency = $\frac{1}{T}$ (Hertz)

The advantage of sinusoidal steady-state is twofold. On one hand, it allows the analysis of a circuit to be reduced to a “static” proportional problem, because one can work with phasors instead of time-dependent functions; on the other hand, it represents the foundation upon which more general analysis tools such as the Fourier transform are built. For these reasons, in electrical and electronic engineering, the study of sinusoidal steady-state is an essential step.

The transition from the time domain to the phasor domain was formalized by Steinmetz, one of the great pioneers of electrical engineering between the 19th and 20th centuries. The Steinmetz transform is simply a way to represent a sinusoidal function as a complex number, in which the amplitude and phase of the signal are encoded respectively in the magnitude and the argument.

If we consider the sinusoidal function $a(t) = A_M \cos(\omega t + \alpha)$, the Steinmetz transform (which we denote by the symbol $\mathcal{S}$) associates it with a complex number A having magnitude A_M and argument α .

$$\mathcal{S}\{a(t)\} = \mathbf{A} = A_M e^{j\alpha} = A_M (\cos \alpha + j \sin \alpha) \quad \text{where } A \text{ is the phasor of } a(t).$$

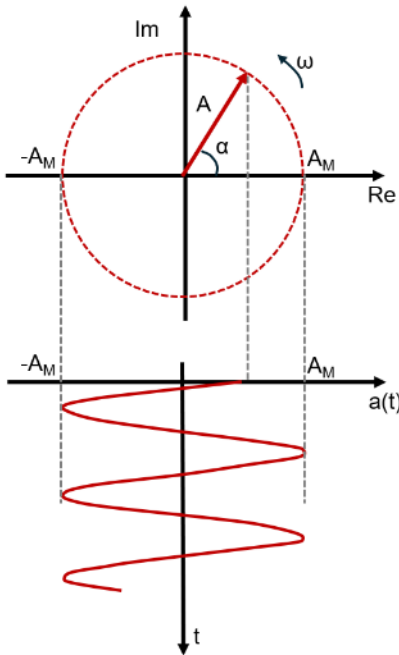
In the last step, Euler’s formulas have been applied:

$$e^{j\alpha} = \cos(\alpha) + j \sin(\alpha)$$

Let us try to give a geometric interpretation of what we have obtained. At time $t = 0$, the phasor A forms an angle α with the horizontal axis (real axis), so its projections onto the real and imaginary axes are respectively $A_M \cos \alpha$ and $A_M \sin \alpha$. Now, let us imagine rotating the phasor with angular velocity ω . At any instant, the projections will be $A_M \cos(\alpha + \omega t)$ and $A_M \sin(\alpha + \omega t)$, thus represented by the complex number s that varies over time according to the relation:

$$s(t) = A_M (\cos(\alpha + \omega t) + j \sin(\alpha + \omega t)) = A_M e^{j(\omega t + \alpha)}$$

Therefore, the original function $a(t)$ is nothing but the real part of this complex number. Recall that α does not vary, and the rotation is only related to ωt .



$$s(t) = \mathbf{A} e^{j\omega t} = A_M e^{j\alpha} e^{j\omega t} = A_M e^{j(\omega t + \alpha)}$$

$$a(t) = \text{Re} [\mathbf{A} e^{j\omega t}] = A_M \cos (\omega t + \alpha)$$

We observe that, by using the transform, the explicit dependence on time disappears, and only the information about phase and amplitude remains, while the frequency ω is assumed as a common (and implicit) context for the entire circuit.

$$\mathcal{S}\{ a(t) \} = A_M e^{j\alpha}$$

The use of this transform greatly simplifies calculations. Instead of manipulating differential equations in time, it becomes possible to treat a circuit as if it were governed by algebraic laws: Ohm's law and Kirchhoff's current and voltage laws remain formally identical, but the quantities are no longer real numbers, rather complex numbers. This is the true strength of the Steinmetz transform: it converts a dynamic problem into a static one, provided one accepts working in the phasor domain.

The properties of this transform are well known and very useful.

- **Linearity**

$$\mathcal{S}\{ k a(t) + h b(t) \} = k \mathcal{S}\{ a(t) \} + h \mathcal{S}\{ b(t) \} \quad \forall h, k \in \mathbb{R}$$

- **Differentiation**

The time derivative of a quantity corresponds, in the phasor domain, to a simple multiplication by $j\omega$.

$$\mathcal{S} \left\{ \frac{d^n a(t)}{dt^n} \right\} = (j\omega)^n \mathcal{S} \{ a(t) \} \quad n: \text{order of the derivative}$$

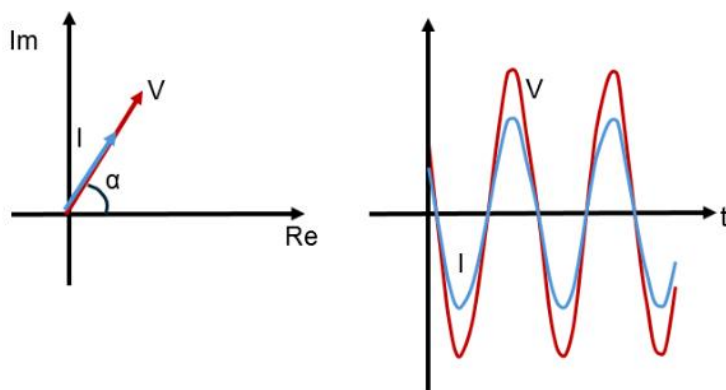
- **Integration**

An integration in time corresponds, in the phasor domain, to a division by $j\omega$.

$$\mathcal{S} \left\{ \int a(t) dt \right\} = \frac{1}{j\omega} \mathcal{S} \{ a(t) \}$$

This direct relationship between differential operations and algebraic factors makes the transition from the physical behavior of circuit elements to their phasor representation immediate.

A resistor, which in steady-state dissipates energy proportionally to the current, maintains exactly the same behavior in sinusoidal steady-state: its impedance is a real number equal to the resistance value R .



$$v(t) = R i(t) = R I_M \cos(\omega t + \alpha_i)$$

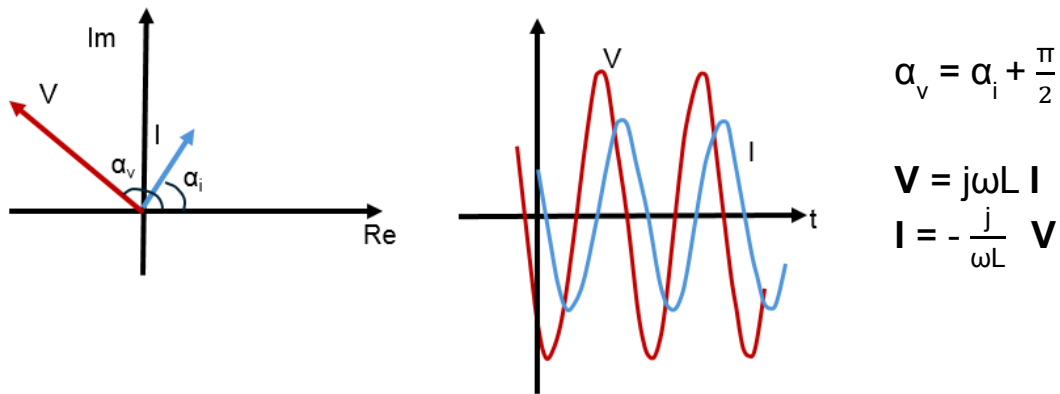
$$i(t) = G v(t) = G V_M \cos(\omega t + \alpha_v)$$

$$\alpha_v = \alpha_i = \alpha$$

$$\mathbf{V} = R \mathbf{I}$$

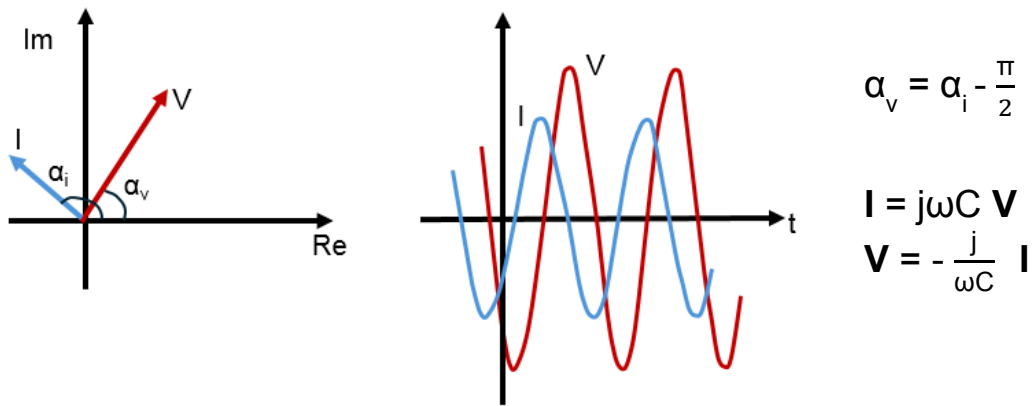
$$\mathbf{I} = G \mathbf{V}$$

An inductor, whose voltage is proportional to the derivative of the current, becomes a two-terminal element with impedance $j\omega L$.



$$v(t) = L \frac{di(t)}{dt} = -\omega L I_M \sin(\omega t + \alpha_i) = \omega L I_M \cos(\omega t + \alpha_i + \frac{\pi}{2})$$

A capacitor, which relates current to the derivative of voltage, is described by an impedance equal to $\frac{1}{j\omega C}$.



$$i(t) = C \frac{dv(t)}{dt} = -\omega C V_M \sin(\omega t + \alpha_v) = \omega C V_M \cos(\omega t + \alpha_v + \frac{\pi}{2})$$

The following table summarizes what has been discussed.

Component	Z (impedance)	Y (admittance)
Resistor	R	G
Inductor	$j\omega L$	$-j/\omega L$
Capacitor	$-j/\omega C$	$j\omega C$

The usefulness of this formalism lies in the fact that, once the complex representation is accepted, circuits in sinusoidal steady-state are analyzed exactly like DC circuits. Resistances are replaced by impedances, Kirchhoff's laws remain unchanged, and the resulting equations are algebraic, solvable with the same well-known techniques: voltage dividers, mesh analysis, nodal analysis. Only at the end, when returning to the time domain, is the physical quantity reconstructed from the phasor using the corresponding cosine or sine.

Transmission Lines

A transmission line is a physical system made up of at least two parallel conductors, one for the forward path and one for the return path, used to carry an electrical signal from one point to another. At first glance, it might seem like a simple wire, not very different from those used in low-frequency circuits. And indeed, when the lengths are small compared to the wavelength of the signal, a wire can be treated as a lumped resistance, possibly with some parasitic inductance. However, as soon as the conductor length becomes comparable to the signal wavelength, this model breaks down: suddenly, the wire is no longer a negligible detail, but becomes an active element of the system that affects how the signal propagates.

It is precisely under these conditions that we speak of a transmission line. The cable is no longer considered a simple ohmic connection, but a medium capable of guiding an electromagnetic wave. In fact, the signal propagating along the line is not a flow of electrons traveling from one end to the other, but rather a wave of electric and magnetic fields confined in the space between the conductors. This concept is easier to understand if we think of radio waves or light: they do not travel by transporting matter, but as electromagnetic energy propagating through space. The transmission line, with its geometry and dielectric, thus becomes a sort of "waveguide" that directs and supports signal propagation.

A transmission line is a distributed-parameter system because its electrical properties, resistance (r), inductance (l), capacitance (c), and conductance (g), cannot be concentrated at a single point or represented by a single two-terminal element, as in low-frequency circuits. Every infinitesimal segment " dx " of the line contributes a small portion of resistance (dr), inductance (dl), capacitance (dc),

and conductance (dg), distributed along its entire length. In other words, every millimeter of cable has its own small inductance (due to the magnetic field around the conductors), its own capacitance (due to the electric field between conductors and dielectric), its own resistance (due to the material opposing current flow), and its own conductance (due to dielectric losses).

The correct model of a transmission line, therefore, is not a single lumped circuit but an infinite succession of elementary cells, each composed of these four parameters. It is as if the line were a continuous mosaic of small r – l – c – g circuits connected one after the other.

Since the line length is not known a priori, r , l , c , and g are defined “per unit length.” For example, r is measured in Ω/m .

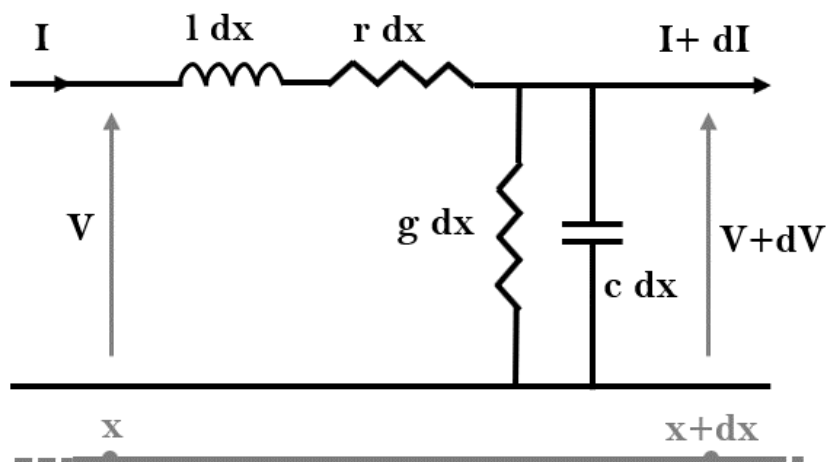


Figure 2: Model of the infinitesimal segment “ dx ” of the line

This distributed nature is the reason why transmission lines become crucial at high frequencies and in communications. At low frequencies, a wire can be approximated as a resistance or at most a lumped inductance, because signal variations are so slow that the entire cable “moves in unison.” But when frequency increases, each point along the line experiences a different phase of the wave, and energy propagates progressively. Only by treating it as a distributed system is it possible to predict reflections, attenuation, and phase shifts, precisely the phenomena that TDR helps us measure.



Figure 3: Example of a transmission line – coaxial cable

Let us therefore consider a generic infinitesimal segment of transmission line of length dx between two cross-sections located at distances x and $x + dx$. Suppose that at $x = 0$ a sinusoidal generator with angular frequency ω is connected. This infinitesimal segment can be represented as shown previously, where r , l , c , and g represent resistance, inductance, capacitance, and conductance per unit length. Let $V(x)$ and $I(x)$ be the phasors of the voltage difference between the two conductors and the current in the line, respectively, both evaluated at distance x . At distance $x + dx$, these quantities become $V(x) + dV(x)$ and $I(x) + dI(x)$.

The voltage increment $dV(x)$ is generated by the current $I(x)$ flowing through the $rdx + j \omega l dx$, thus:

$$V(x) - [V(x) + dV(x)] = dV(x) = (r dx + j \omega l dx) I(x)$$

The current increment $dI(x)$ is due to the current flowing through the admittance $(g dx + j \omega c dx)$:

$$I(x) - [I(x) - dI(x)] = dI(x) = (g dx + j \omega c dx) V(x)$$

Dividing the above equations by dx , we obtain:

$$\frac{dV(x)}{dx} = -(r + j \omega l) I(x)$$

$$\frac{dI(x)}{dx} = -(g + j \omega c) V(x)$$

These relations are known as the telegrapher's equations. Differentiating the first with respect to x and substituting the derivative of $I(x)$ from the second, we obtain:

$$\frac{d^2 V(x)}{dx^2} = \gamma^2 V(x)$$

The coefficient

$$\gamma = \sqrt{(r + j\omega l)(g + j\omega c)}$$

is called the propagation constant.

Since γ is a complex quantity, it can be written as $\gamma = \alpha + j\beta$, where α is the attenuation constant and β is the phase constant.

The general solution of the differential equation

$$\frac{d^2 V(x)}{dx^2} = \gamma^2 V(x)$$

is:

$$V(x) = V_1 e^{-\gamma x} + V_2 e^{\gamma x}$$

where V_1 and V_2 are complex constants determined by boundary conditions.

We observe that $V_1 e^{-\gamma x}$ and $V_2 e^{\gamma x}$ are two waves traveling respectively toward $x+$ (toward the load) and $x-$ (toward the source).

Returning to the voltage derivative equation and computing $I(x)$:

$$\frac{dV(x)}{dx} = -(r + j\omega l) I(x) \rightarrow I(x) = -\frac{1}{r + j\omega l} \frac{dV(x)}{dx}$$

Substituting the expression for $V(x)$:

$$\begin{aligned} I(x) &= -\frac{1}{r + j\omega l} \frac{dV(x)}{dx} = -\frac{1}{r + j\omega l} (-\gamma V_1 e^{-\gamma x} + \gamma V_2 e^{\gamma x}) \\ &= \frac{\gamma}{r + j\omega l} (V_1 e^{-\gamma x} - V_2 e^{\gamma x}) \\ &= \frac{\sqrt{(r + j\omega l)(g + j\omega c)}}{r + j\omega l} (V_1 e^{-\gamma x} - V_2 e^{\gamma x}) \\ &= \sqrt{\frac{1}{\frac{r + j\omega l}{g + j\omega c}}} (V_1 e^{-\gamma x} - V_2 e^{\gamma x}) = \frac{V_1}{Z_0} e^{-\gamma x} - \frac{V_2}{Z_0} e^{\gamma x} \end{aligned}$$

where:

$$Z_0 = \sqrt{\frac{r + j\omega l}{g + j\omega c}}$$

This quantity has the dimensions of impedance and is called the characteristic impedance of the line.

In practice, it is useful to characterize the line starting from the load, so we perform a change of reference by defining $d = (L - x)$, and rewrite:

$$\begin{aligned} V(x) &= V_1 e^{-\gamma x} + V_2 e^{\gamma x} \rightarrow V(d) = V_1 e^{-\gamma(L-d)} + V_2 e^{\gamma(L-d)} \\ &= V_1 e^{-\gamma L} e^{\gamma d} + V_2 e^{\gamma L} e^{-\gamma d} \\ I(x) &= \frac{V_1}{Z_0} e^{-\gamma x} - \frac{V_2}{Z_0} e^{\gamma x} \rightarrow I(d) = \frac{V_1}{Z_0} e^{-\gamma(L-d)} - \frac{V_2}{Z_0} e^{\gamma(L-d)} \\ &= \frac{V_1}{Z_0} e^{-\gamma L} e^{\gamma d} - \frac{V_2}{Z_0} e^{\gamma L} e^{-\gamma d} \end{aligned}$$

At $d = 0$ (i.e., at the load, $x = L$):

$$\begin{aligned} V(0) &= V_1 e^{-\gamma L} + V_2 e^{\gamma L} \\ I(0) &= \frac{V_1}{Z_0} e^{-\gamma L} - \frac{V_2}{Z_0} e^{\gamma L} \end{aligned}$$

From these expressions, the term $V_1 e^{-\gamma L}$ represents the voltage at the load due to the forward wave, while $V_2 e^{\gamma L}$ represents the voltage due to the reflected wave. The same applies to currents. The term $\frac{V_1}{Z_0} e^{-\gamma L}$ represents the current at the load due to the forward wave, while $-\frac{V_2}{Z_0} e^{\gamma L}$ represents the current due to the reflected wave

Let:

$$V_{dL} = V_1 e^{-\gamma L}$$

$$V_{rL} = V_2 e^{\gamma L}$$

$$I_{dL} = \frac{V_1}{Z_0} e^{-\gamma L} = \frac{V_{dL}}{Z_0}$$

$$I_{rL} = -\frac{V_2}{Z_0} e^{\gamma L} = -\frac{V_{rL}}{Z_0}$$

Thus:

$$V(d) = V_1 e^{-\gamma L} e^{\gamma d} + V_2 e^{\gamma L} e^{-\gamma d} = V_{dL} e^{\gamma d} + V_{rL} e^{-\gamma d}$$

$$I(x) = \frac{V_1}{Z_0} e^{-\gamma L} e^{\gamma d} - \frac{V_2}{Z_0} e^{\gamma L} e^{-\gamma d} = I_{dL} e^{\gamma d} + I_{rL} e^{-\gamma d}$$

The voltage reflection coefficient at the load is defined as:

$$\Gamma_L = \frac{V_{rL}}{V_{dL}} = -\frac{I_{rL}}{I_{dL}}$$

Rewriting:

$$V(d) = V_{dL} e^{\gamma d} + V_{rL} e^{-\gamma d} = V_{dL} \left(e^{\gamma d} + \frac{V_{rL}}{V_{dL}} e^{-\gamma d} \right) = V_{dL} (e^{\gamma d} + \Gamma_L e^{-\gamma d})$$

$$\begin{aligned} I(d) &= I_{dL} e^{\gamma d} + I_{rL} e^{-\gamma d} = I_{dL} \left(e^{\gamma d} + \frac{I_{rL}}{I_{dL}} e^{-\gamma d} \right) = I_{dL} (e^{\gamma d} - \Gamma_L e^{-\gamma d}) = \\ &= \frac{V_{dL}}{Z_0} (e^{\gamma d} - \Gamma_L e^{-\gamma d}) \end{aligned}$$

At $d = 0$ (on load):

$$V(0) = V_{dL} (e^{\gamma_0} + \Gamma_L e^{-\gamma_0}) = V_{dL} (1 + \Gamma_L)$$

$$I(0) = \frac{V_{dL}}{Z_0} (1 - \Gamma_L)$$

Thus:

$$\frac{V(0)}{I(0)} = Z_L = \frac{V_{dL} (1 + \Gamma_L)}{\frac{V_{dL}}{Z_0} (1 - \Gamma_L)} = Z_0 \frac{(1 + \Gamma_L)}{(1 - \Gamma_L)}$$

From which:

$$\Gamma_L = \frac{(Z_L - Z_0)}{(Z_L + Z_0)}$$

We observe that:

- $|\Gamma_L|$ represents the amplitude ratio between reflected and incident waves;
- $\angle \Gamma_L$ (phase) indicates the phase shift introduced by reflection. For complex loads Γ_L is a complex number so the reflected wave can change phase with respect to the incident one.
- Notable cases:
 - If $Z_L \rightarrow \infty$, then $\Gamma_L = 1 \rightarrow V_{dL} = V_{rL}$ e $I_{dL} = -I_{rL}$
 - If $Z_L \rightarrow 0$, then $\Gamma_L = -1 \rightarrow V_{dL} = -V_{rL}$ e $I_{dL} = I_{rL}$
 - If $Z_L = Z_0$, then $\Gamma_L = 0 \rightarrow$ no reflection (perfect matching).
 - If Z_L is purely reactive ($Z_L = jX$, $X \in \mathbb{R}$), then $|\Gamma_L|=1 \rightarrow$ total reflection with phase shift.

When Z_L and Z_0 are complex numbers (case of reactive loads or lossy lines), Γ_L is complex:

$$\Gamma_L = \frac{(Z_L - Z_0)}{(Z_L + Z_0)} = \frac{(R_{Load} - R_0) + j(X_{Load} - X_0)}{(R_{Load} + R_0) + j(X_{Load} + X_0)}$$

$$|\Gamma_L| = \frac{|Z_L - Z_0|}{|Z_L + Z_0|} \quad \arg(\Gamma_L) = \arg(Z_L - Z_0) - \arg(Z_L + Z_0)$$

The magnitude indicates how much power is reflected; the phase determines inversion or phase shift.

A common RF standard is the Voltage Standing Wave Ratio (VSWR), defined as:

$$VSWR = \frac{1 + |\Gamma_L|}{1 - |\Gamma_L|}$$

- VSWR = 1 corresponds to perfect matching, ($|\Gamma_L| = 0$),
- VSWR $\rightarrow \infty$ is obtained for $|\Gamma_L| \rightarrow 1$ (open or short Z_L , or purely reactive Z_L).

We have seen that, in general, two waves propagate along the line: a forward wave and a reflected wave.

But, at what speed do they travel?

The propagation velocity of waves in a transmission line is a key aspect when studying phenomena such as Time Domain Reflectometry (TDR). Electromagnetic waves traveling along a transmission line do not propagate at the speed of light in vacuum ($c \approx 3 \cdot 10^8$ m/s), but more slowly, depending on the dielectric between the conductors.

Mathematically:

$$v = \frac{c}{\sqrt{\epsilon_r}}$$

Where:

c : speed of light in vacuum ($3 \cdot 10^8$ m/s)

ϵ_r : is the relative dielectric constant of the material that fills the line.

For example, in common coaxial cables with solid polyethylene dielectric $\epsilon_r \approx 2.25$,

so, the propagation velocity is $v = \frac{c}{\sqrt{\epsilon_r}} = \frac{3 \cdot 10^8}{\sqrt{2.25}} \approx 2 \cdot 10^8$ m/s

The ratio $\frac{v}{c} = \frac{1}{\sqrt{\epsilon_r}}$ is called the velocity factor (VF). A typical value is 0.66 (e.g., RG-58) with $\epsilon_r=2.3$.

Estimating propagation speed is essential to:

- calculate the propagation delay of a signal (the time it takes to travel a length of line),
- estimate the distance of a defect with TDR: in fact, measuring the pulse's round-trip time, multiplied by the propagation speed, directly gives the location of the discontinuity.

3. TDR Study: Bouncing Diagram

In the previous chapter, we examined how a transmission line behaves from an electronic and mathematical standpoint. Let us now try to provide an intuitive explanation of the results obtained.

Imagine having a taut string. If we tug at one end, we'll see a wave travel along the string until it reaches the other end. If the end is free, the wave "bounces" and comes back; if it's attached to a wall, it bounces, but inverted. This is a very simple way to understand what happens in electrical cables: they behave like "strings for electrical signals".

A transmission line is nothing more than an electrical cable (for example, a coaxial cable or a twisted pair) in which a signal does not spread instantaneously, but travels at a certain speed that depends on the insulating material and the geometry of the cable. Just like the wave on the rope, the electrical signal also needs time to travel its path.

As the signal propagates, we can think of it as a forward wave that starts from the generator and travels toward the load (that is, toward the device connected to the cable). However, if the load is not perfectly "matched" to the cable, part of the energy cannot enter and is sent back: this gives rise to a reflected wave.

If the load is exactly equal to the characteristic impedance of the line, the entire forward wave is absorbed: no reflection occurs.

If instead the load is too large (open circuit) or too small (short circuit), the wave returns as a total reflection.

In intermediate cases (for example, a resistive load different from the ideal value), only part of the wave is absorbed and the rest is reflected.

This is why, in a real transmission line, there are always two waves:

- the forward wave, which carries the signal from the generator to the load;
- the reflected wave, which returns when there is a mismatch.

The superposition of these waves can create observable phenomena, such as voltage variations along the cable or even standing waves, just like on a stretched rope vibrating at multiple points.

A transmission line is a distributed system, not a single “component”: resistance, inductance, capacitance, and conductance are distributed along its entire length.

Let us now examine the key quantities that define the behavior of a transmission line and are essential for understanding TDR analysis:

- Characteristic impedance Z_0 , which for a low-loss line can be expressed as $Z_0 = \sqrt{\frac{l}{c}}$ where “ l ” and “ c ” are the inductance and capacitance per unit length. Z_0 represents a sort of “natural impedance” of the line, determined exclusively by its geometry and the dielectric that fills it. It does not depend on the cable length: a 50 Ω coaxial cable remains such whether it is one meter long or one hundred meters. TDR is based precisely on comparing the characteristic impedance of the line with the impedance encountered along its path: whenever there is a difference, a reflection is generated that can be measured.
- The signal does not travel along the line instantaneously but takes a finite time, TD (propagation delay), which depends on the length of the line and the propagation speed in the insulating medium.

The propagation speed is equal to $v = \frac{c}{\sqrt{\epsilon_r}}$, where “ c ” is the speed of light and ϵ_r is the relative dielectric constant of the line. A signal, therefore, travels at a speed that is a fraction of the speed of light, between 60% and 80%. This data is crucial in TDR because it allows us to convert the round-trip time of the reflected signal into an actual distance along the cable.

- The velocity factor $FV = \frac{v}{c}$, the ratio between the speed of propagation in the line and the speed of light in vacuum. It is always indicated in cable datasheets and is essential when using TDR to calculate the exact location of a defect.
- The reflection coefficient $\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0}$ where Z_L is the load impedance and Z_0 is the characteristic impedance of the line. The reflection coefficient Γ_L is a dimensionless number that quantifies the ratio between the reflected wave and the incident wave at a discontinuity.
- Spatial resolution, or the ability to distinguish two discontinuities or two defects along a transmission line. It is the minimum distance that must separate two events for them to be observed as distinct in the time domain. In mathematical terms, $\Delta z = \frac{v \Delta T}{2}$ where:
 v : velocity of propagation in the medium
 ΔT : the effective temporal resolution, i.e., the pulse duration or the rise time of the step.
 $\frac{1}{2}$: factor that takes into account the fact that the signal makes a "round trip" to generate the reflection.

For example, if the test pulse lasts $\Delta T = 0.2\text{ns}$ and $v = 2 \cdot 10^8 \text{ m/s}$, then $\Delta z = \frac{v \Delta T}{2} = \frac{2 \cdot 10^8 \cdot 0.2 \cdot 10^{-9}}{2} = 2 \text{ cm}$, so I can detect defects further than 2 cm away. The same result is obtained with a step where the rise time is 0.2 ns.

Bouncing diagram

The bouncing diagram, or bounce diagram, is a very useful graphical tool for intuitively understanding how signals propagate and reflect along a transmission line. It is particularly valuable in the study of TDR because it allows us to predict the waveform that will be observed at the source or load.

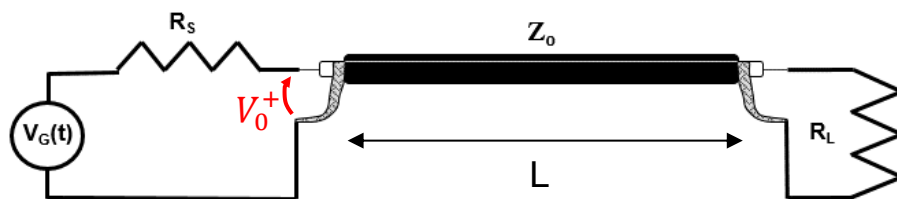
Let's look step by step at how it is constructed and what it represents.

When a signal (for example, a voltage pulse) enters a transmission line, it travels at a certain speed to the load. If the load is perfectly matched, the signal stops there: there are no reflections. If, however, the load does not match the characteristic impedance of the line, part of the signal returns as a reflected wave.

This reflected wave returns to the source and, if the source is also not matched, it is in turn partially reflected again, and so on. This creates a "ping-pong" of back and forth waves, which over time add their effects at the ends of the line.

The reflection (bouncing) diagram is a way to graphically represent all these journeys and reflections on the time-space plane.

Let's analyze the following circuit.



Let's start by calculating the signal injected into the line at time $t=0$, or, applying Ohm's law:

$$V_0^+ = V_G \frac{Z_0}{Z_0 + R_s}$$

This signal, after a propagation time $T = \frac{L}{v_p}$, will reach the load. Recall that v_p is the wave velocity in the transmission medium (coaxial cable) and L is the length of the conductor.

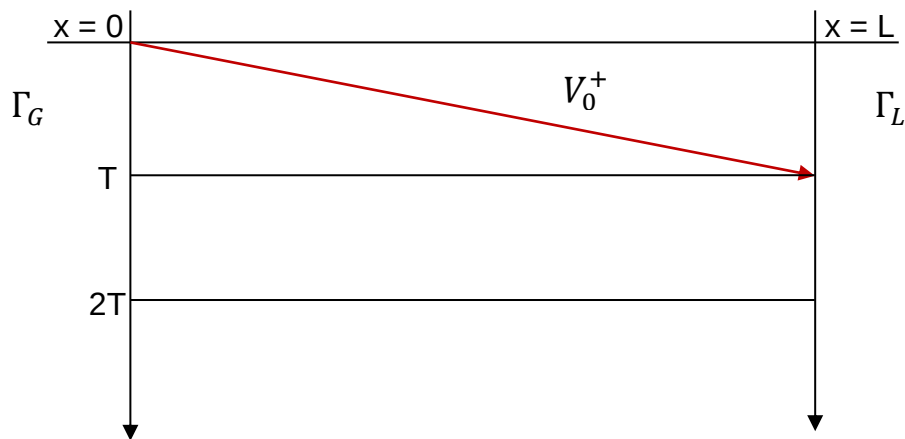
The bouncing diagram is typically represented on a two-axis plane:

- Horizontal axis: represents space. At the top left is the position of the source and at the right is the load. The horizontal line connecting them represents the transmission line.
- Vertical axis: represents time. Zero is marked when the generator sends the first wave.

On this plane, forward waves are drawn as diagonal lines inclined from the generator to the load with a slope proportional to the propagation velocity; reflected waves are drawn as diagonal lines inclined from the load toward the generator.

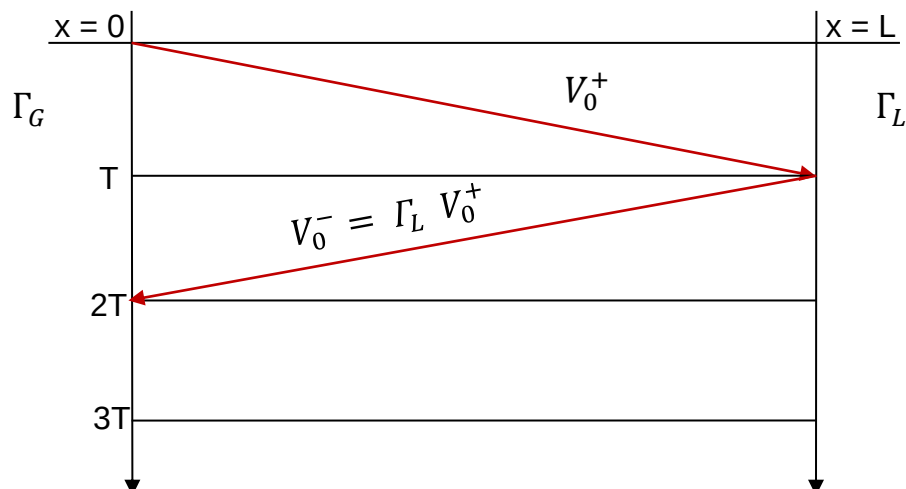
Each reflection is accompanied by a change in sign and/or amplitude of the wave depending on the reflection coefficient Γ at the point of discontinuity (load or generator).

In this case, we can immediately represent the forward wave V_0^+ traveling from the generator ($x=0$) to the load ($x=L$).

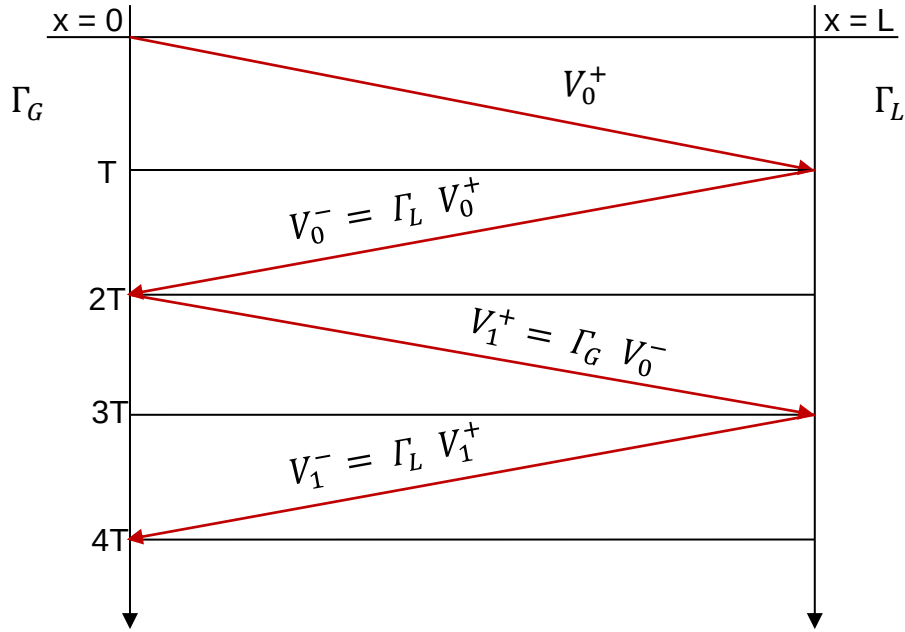


On the right and left of the graph, it is appropriate to report the reflection coefficients of the load Γ_L and the generator Γ_G , respectively.

As soon as the wave reaches the load, it is reflected (if there is no perfect match), producing a reflected wave $V_0^- = \Gamma_L V_0^+$.



When V_0^- reaches the generator, it is partially reflected, generating $V_1^+ = \Gamma_G V_0^-$. This wave heads towards the load and the mechanism repeats itself iteratively, creating a “ping-pong” diagram.



In general, the algorithm involves iteratively applying the following relationships:

$$V_n^- = \Gamma_L V_n^+$$

$$V_{n+1}^- = \Gamma_G V_n^- = \Gamma_G \Gamma_L V_n^+$$

In practice, the signal we inject into the transmission line (coaxial cable) bounces between the generator and the load, which is appropriately modulated by Γ_L and Γ_G .

Considering the constant value of the driving step and the various subsequent "bounces" $V_{n+1}^- = V_n^+ \Gamma_G \Gamma_L$, the voltage at point x of the line under steady-state conditions, i.e., at infinite time, can be written as:

$$\begin{aligned}
V(x) &= V_{0,x}^+ (1 + \Gamma_L + \Gamma_L \Gamma_G + \Gamma_L^2 \Gamma_G + \Gamma_L^2 \Gamma_G^2 + \dots) \\
&= V_0^+ (1 + \Gamma_L \Gamma_G + (\Gamma_L \Gamma_G)^2 + \dots) \\
&+ V_0^+ \Gamma_L ((1 + \Gamma_L \Gamma_G + (\Gamma_L \Gamma_G)^2 + \dots))
\end{aligned}$$

Since $\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$, we can write:

$$V(x) = V_{0,x}^+ \left(\frac{1}{1 - \Gamma_L \Gamma_G} + \frac{\Gamma_L}{1 - \Gamma_L \Gamma_G} \right) = V_{0,x}^+ \left(\frac{1 + \Gamma_L}{1 - \Gamma_L \Gamma_G} \right)$$

This is the voltage that develops on the line under steady-state conditions (for $t \rightarrow \infty$).

Let's look at the following figure: it shows the bounce diagram for a single transmission line with load reflection coefficient Γ_L and generator reflection coefficient Γ_G . The voltage at a point on the transmission line at time t is the sum of all reflections that have occurred up to the specified time. Therefore, if we want to plot the voltage at a point on the line as a function of time, we draw a vertical line on the bounce diagram at the indicated location. The reflected signal (or pulse) passes through that point every time the vertical line crosses the line representing the rising edge of the reflected pulse. At a given instant, the voltage on the transmission line is the sum of all reflection amplitudes below the current instant.

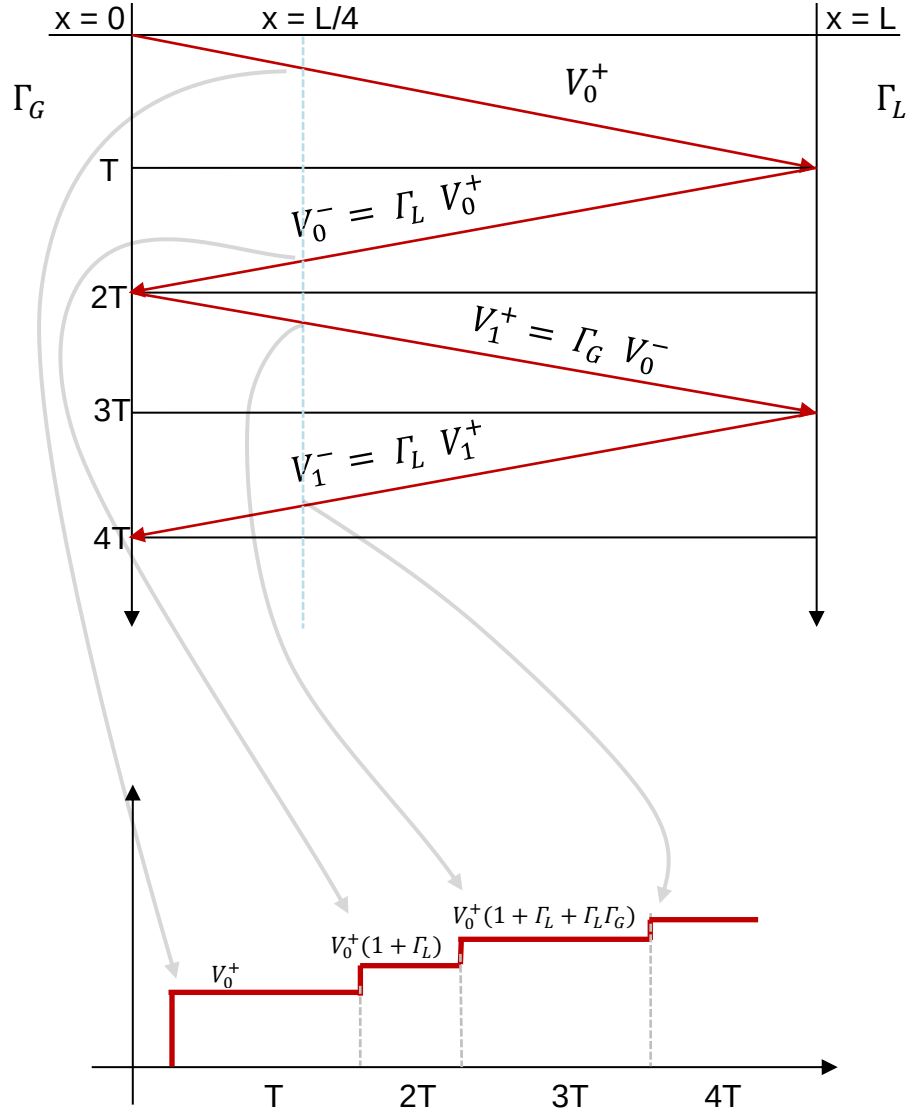


Figure 4: Voltage as a function of time at position $L/4$ obtained from the bouncing diagram.

If we put ourselves on the load we have that $V_{0,L}^+ = V_0^+ = V_G \frac{Z_0}{Z_0 + R_s}$, so:

$$V_{L,\infty} = V_0^+ \left(\frac{1 + \Gamma_L}{1 - \Gamma_L \Gamma_G} \right) = V_G \frac{Z_0}{Z_0 + R_s} \left(\frac{1 + \frac{Z_L - Z_0}{Z_L + Z_0}}{1 - \frac{Z_L - Z_0}{Z_L + Z_0} \cdot \frac{Z_G - Z_0}{Z_G + Z_0}} \right) = V_G \frac{R_L}{R_L + R_s}$$

We have omitted the mathematical calculations, but it is important to note that the voltage on the load, once the transient is over, is equal to:

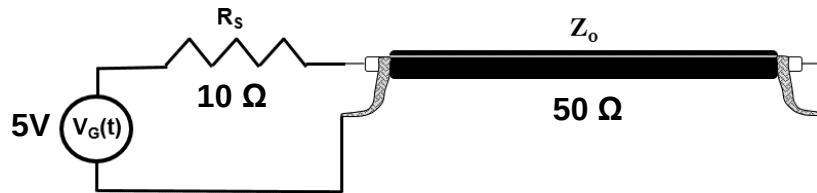
$$V_{L,\infty} = V_G \frac{R_L}{R_L + R_S}$$

That is, it does not depend on the transmission line (Z_0).

Now we can examine some edge cases that will help us better understand the TDR technique.

First case: open line, no load.

Let's begin by examining the case where no load is connected to the transmission line.



Let's start by calculating the propagation time T . Suppose we have a 20-centimeter-long coaxial cable with $\epsilon_r = 2.25$.

We know that $v = \frac{c}{\sqrt{\epsilon_r}} = \frac{3 \cdot 10^8}{\sqrt{2.25}} = 2 \cdot 10^8 \frac{m}{s}$, so $T = \frac{l}{v} = \frac{0.2}{2 \cdot 10^8} = 1ns$

The voltage V_0^+ is calculated using the voltage divider formula (Ohm's law):

$$V_0^+ = V_G \frac{Z_0}{Z_0 + R_S} = 5 \frac{50}{50 + 10} = 4.16 V$$

Let's calculate the reflection coefficients at the source (Generator) and the load (Load).

$$\Gamma_G = \frac{Z_S - Z_0}{Z_S + Z_0} = \frac{10 - 50}{10 + 50} = -\frac{4}{6} = -\frac{2}{3}$$

$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{\frac{Z_L}{Z_L} - \frac{Z_0}{Z_L}}{\frac{Z_L}{Z_L} + \frac{Z_0}{Z_L}} = \frac{1 - \frac{Z_0}{Z_L}}{1 + \frac{Z_0}{Z_L}} = \frac{1 - \frac{Z_0}{\infty}}{1 + \frac{Z_0}{\infty}} = 1$$

As we have already said, in a transmission line, the voltage at a certain point is always the sum of two waves:

- the incident wave V^+ , which travels from the source to the load,
- the reflected wave V^- , which travels from the load to the source.

Therefore, at any point along the line, and in particular at the load, the following applies:

$$V_{Load}(t) = V^+(t) + V^-(t)$$

When a wave front reaches the load, part of the wave is absorbed and part is reflected according to the reflection coefficient Γ_L .

The general rule is:

$$V^- = \Gamma_L \cdot V^+$$

In the case in question:

$$V^- = \Gamma_L \cdot V^+ = 1 \cdot V^+ = V^+$$

That is, every time a wave reaches the load, it generates a reflected wave identical to the incident one.

Immediately after V^+ arrives at the load, the resulting voltage is the sum of:

$$V_{Load} = V^+ + V^- = V^+ + \Gamma_L \cdot V^+ = V^+ + V^+ = 2 \cdot V^+$$

In practice, each arrival at the load produces an increase equal to $2 \cdot V^+$ since its immediate reflection is added to the incident wave.

Let's start filling in the bouncing diagram.

First iteration: n=0 (starting at t=0 and arriving at t = T)

We inject the pulse: $V_0^+ = 4.16V$. This voltage reaches the load (open circuit) and bounces completely. The sum of the incident and reflected wave generates a voltage on the load equal to $2 \cdot V_0^+ = 8.33V$. Meanwhile, the reflected wave heading towards the generator has a voltage $V_0^- = 4.16V$.

Second iteration: n=1 (arriving at t = 3T)

The source sees V_0^- arriving and reflects it according to $V_1^+ = \Gamma_G V_0^- = -\frac{2}{3} \cdot 4.16V = -2.77V$. This voltage is added to the wave arriving from the load; So, at the source we have a voltage $V_1^+ + V_0^- = -2.77 + 4.16 = 1.39V$

The wave V_1^+ (equal to $-2.77V$) arrives at the load and is completely reflected, so $V_1^- = -2.77V$.

At the load we have a resulting voltage equal to $2 \cdot V_1^+ = -5.54V$.

Third iteration: n = 2 (arrival at t = 5 T)

The wave V_1^- arrives at the source.

It is reflected, generating $V_2^+ = \Gamma_G V_1^- = -\frac{2}{3} \cdot -2.77V = 1.84V$.

At the source we have a total voltage equal to $V_1^- + V_2^+ = -2.77 + 1.84 = -0.92V$.

Wave V_2^+ arrives at the load and is completely reflected, resulting in $V_2^- = 1.84V$.

The voltage at the load is $2 \cdot V_2^+ = 2 \cdot 1.84 = 3.704V$.

Fourth iteration: n=3 (arrival at t = 7T)

V_2^- arrives at the source and is reflected, generating $V_3^+ = \Gamma_G V_2^- = -\frac{2}{3} \cdot 1.84 = -1.234V$.

At the source, the voltage is $V_2^- + V_3^+ = 1.84 + (-1.234) = 0.606V$

Wave V_3^+ arrives at the load and is completely reflected: $V_3^- = -1.234V$. The voltage at the load is $2 \cdot V_3^+ = 2 \cdot -1.234 = -2.467V$.

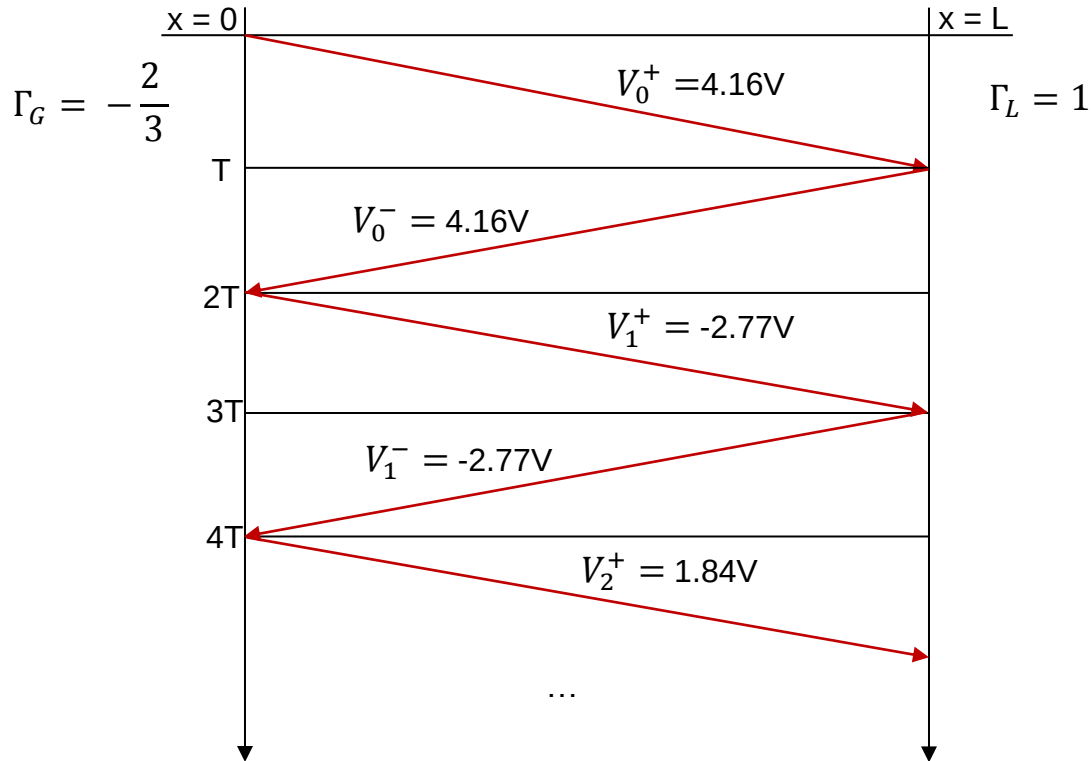


Figure 5: Construction of the bouncing diagram

Let's now see if the simulation with LTSpice confirms the analytical results.

First, we replicate the circuit to be simulated, inserting a step voltage source (5V with a 1ns rise time), a 10 ohm resistor R_s , and a transmission line, namely LTSpice's native T-Line (transmission line) object. This object must be parameterized by specifying the impedance Z_0 (50 ohms) and the propagation delay TD.

Note that, given an observation point, the distance between two successive pulses is equal to $2TD$ (round trip).

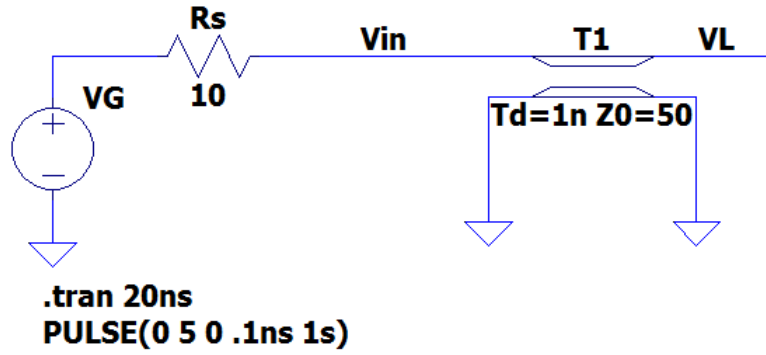


Figure 6: Circuit for simulation "1"

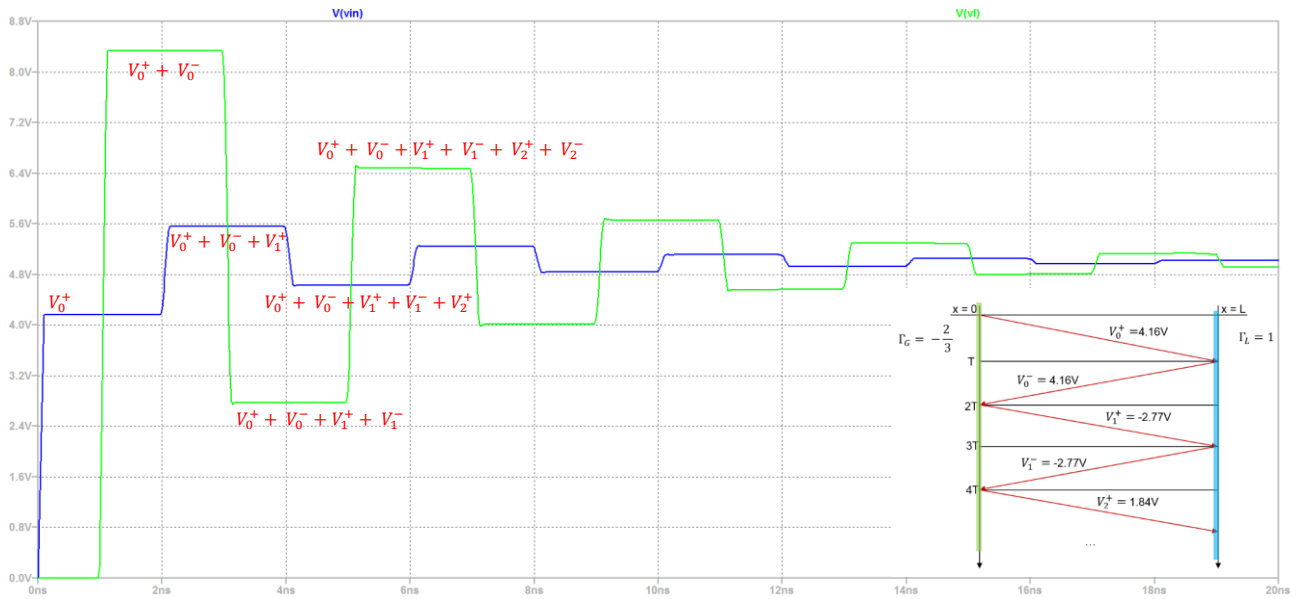


Figure 7: Simulation results "1"

The simulation results are consistent with those obtained previously.

For example, I provide a comparison between the step response (V_{IN} e V_L) and the pulse response (V_{IN2} , V_{L2}). The information content is obviously the same.

Note that in the pulse response, the value of the voltages at infinite time is zero (because the excitation signal lasts 1 ns), while in the step response it settles

$$\text{around } V_{\infty} = V_0^+ \left(\frac{1 + \Gamma_L}{1 - \Gamma_L \Gamma_G} \right) = 4.16 \left(\frac{1 + 1}{1 - \frac{2}{5}} \right) = 4.16 \left(\frac{2}{\frac{3}{5}} \right) = 4.99V$$

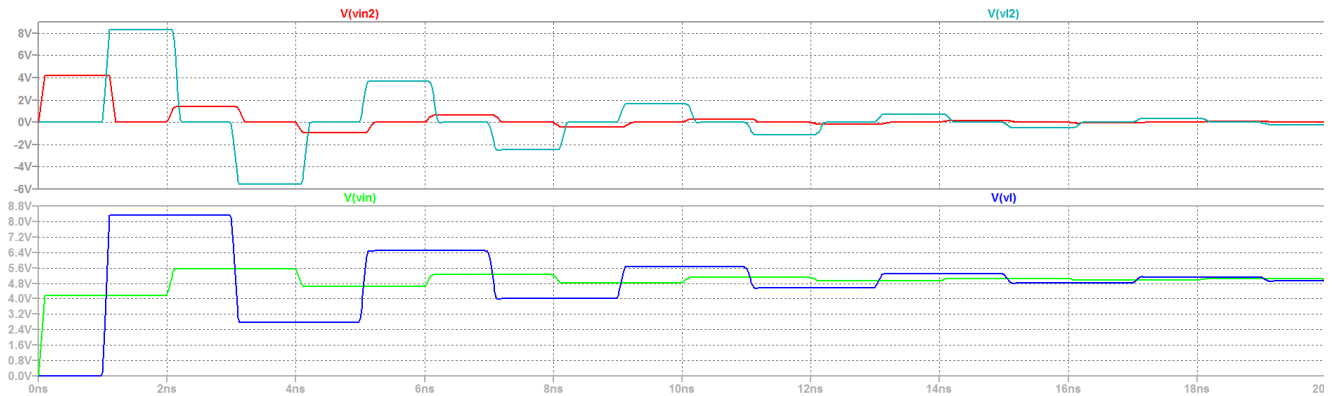
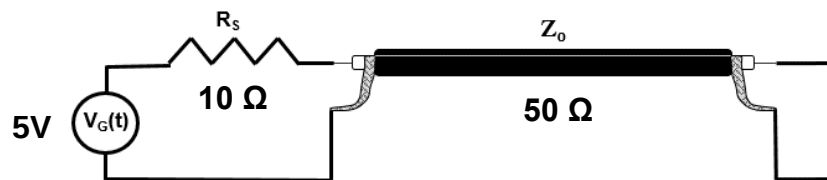


Figure 8: Step and pulse response of circuit "1"

Second case: Short-circuited load

A second interesting case is when the load is short-circuited.



We proceed as in the previous case.

$$V_0^+ = V_G \frac{Z_0}{Z_0 + R_s} = 5 \frac{50}{50 + 10} = 4.16 \text{ V}$$

Let's calculate the reflection coefficients at the source (Generator) and the load (Load).

$$\Gamma_G = \frac{Z_s - Z_0}{Z_s + Z_0} = \frac{10 - 50}{10 + 50} = -\frac{4}{6} = -\frac{2}{3}$$

$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{0 - Z_0}{0 + Z_0} = -1$$

Let's start filling in the bouncing diagram.

First iteration: n=0 (starting at t=0 and arriving at t = T)

$V_0^+ = 4.16V$. This voltage arrives at the load (short circuit) and bounces back with a completely reversed sign.

$V_0^- = -4.16V$. The sum of the incident and reflected waves obviously generates a zero voltage on the load because it is a short circuit.

The current flowing through the load is $I_{Load,0} = \frac{V_0^+ - V_0^-}{Z_0} = \frac{2 V_0^+}{Z_0} = \frac{8.32}{50} = 0.166 A$

Second iteration: n=1 (arriving at t = 3T)

The source sees V_0^- arriving and reflects it according to $V_1^+ = \Gamma_G V_0^- = -\frac{2}{3} \cdot (-4.16V) = 2.77V$. This voltage is added to the wave arriving from the load; therefore, at the source we have a voltage $V_1^+ + V_0^- = 2.77 - 4.16 = -1.39 V$

The wave V_1^+ (pari a $2.77V$) arrives at the load and is completely reflected and reversed in sign, so $V_1^- = -2.77V$.

At the load we have a resulting zero voltage and a current

$$I_{Load,1} = \frac{V_1^+ - V_1^-}{Z_0} = \frac{2.77 + 2.77}{Z_0} = \frac{5.54}{50} = 0.11 A$$

Third iteration: n = 2 (arrival at t = 5T)

The wave V_1^- ($-2.77V$) arrives at the source.

It is reflected, generating $V_2^+ = \Gamma_G V_1^- = -\frac{2}{3} \cdot (-2.77V) = 1.84V$.

At the source, we have a total voltage equal to $V_1^- + V_2^+ = -2.77 + 1.84 = -0.92V$.

The V_2^+ wave reaches the load and is completely bounced, so $V_2^- = -1.84V$.

The voltage at the load is zero and the current:

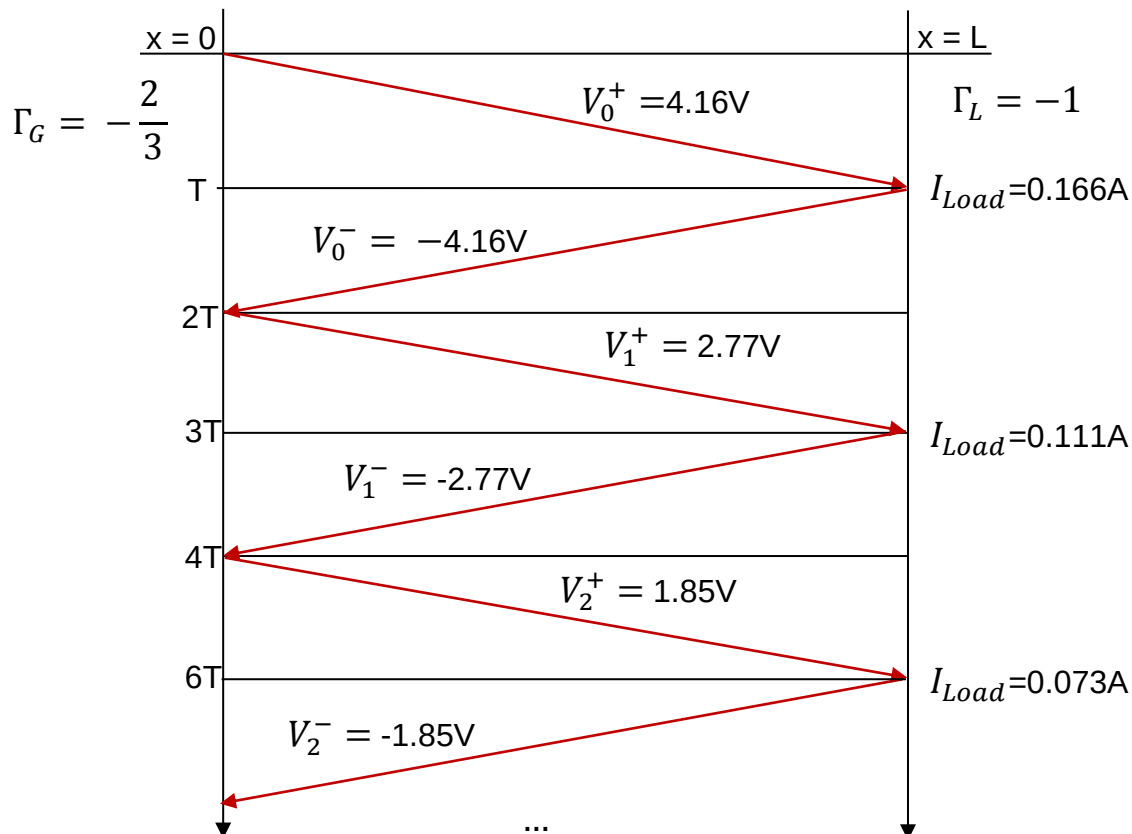
$$I_{Load,2} = \frac{V_2^+ - V_2^-}{Z_0} = \frac{1.84 - (-1.84)}{50} = \frac{5.54}{50} = 0.073 \text{ A}$$

Fourth iteration: n=3 (arrival at t = 7T)

The wave V_2^- arrives at the source and is reflected, generating $V_3^+ = \Gamma_G V_2^- = -\frac{2}{3} \cdot (-1.84) = 1.234 \text{ V}$.

At the source, there is a voltage equal to $V_2^- + V_3^+ = 1.84 + (1.234) = 3.07 \text{ V}$. The V_3^+ wave arrives at the load, is completely bounced and its sign reversed: $V_3^- = -1.234 \text{ V}$. The voltage on the load is zero and the current is:

$$I_{Load,3} = \frac{V_3^+ - V_3^-}{Z_0} = \frac{1.234 - (-1.234)}{50} = \frac{2.468}{50} = 0.049 \text{ A}$$



Let's reproduce the circuit in LTSpice. We simulate the short circuit on the load with a resistor of 0.0001 ohm.

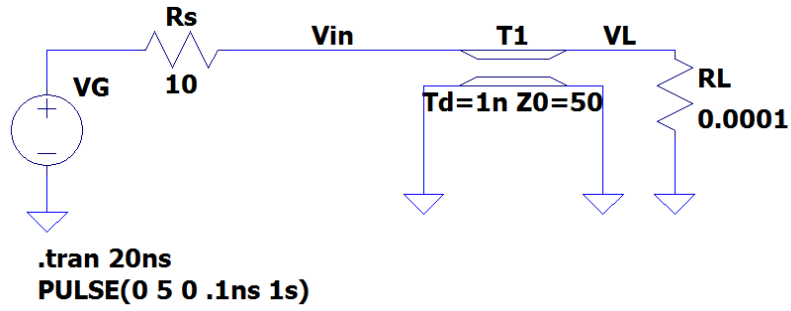


Figura 9: Circuito per simulazione “2”

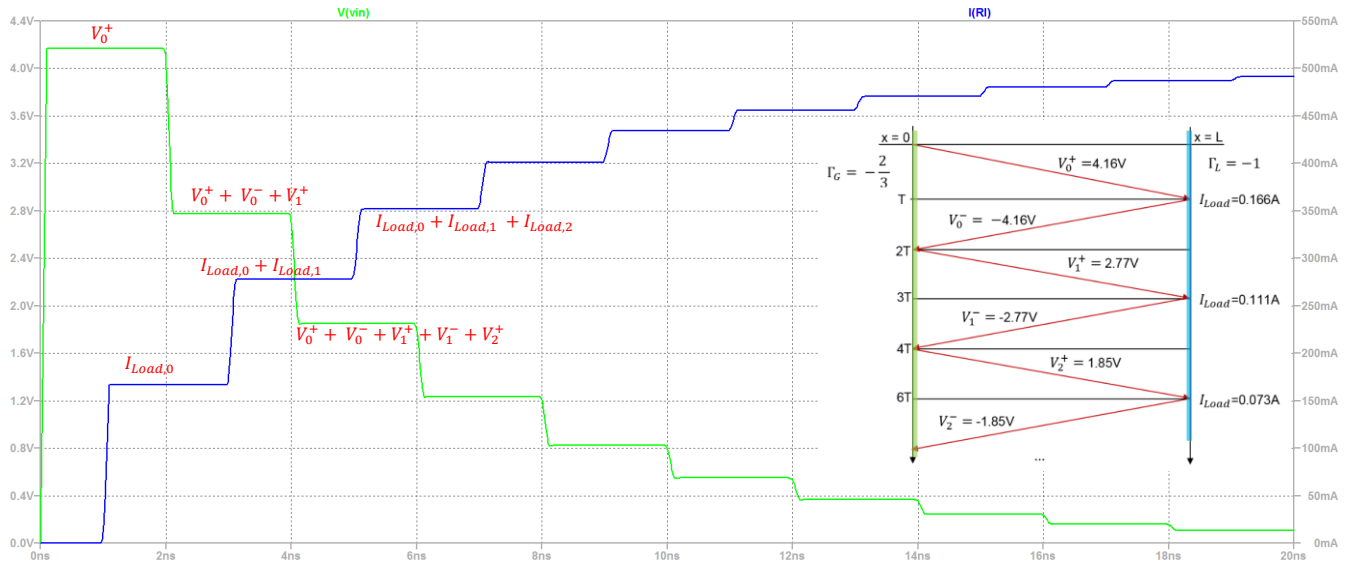


Figure 10: Simulation results “2”

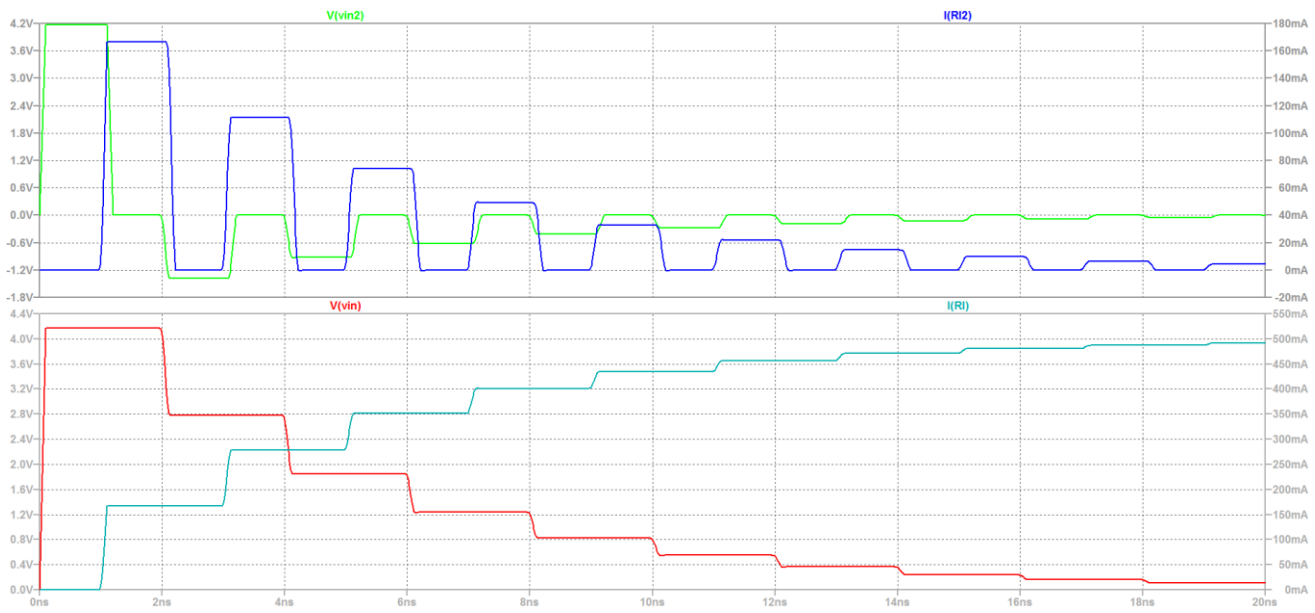
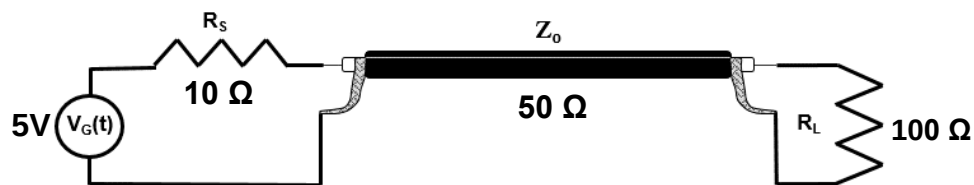


Figure 11: Step and pulse response of circuit “2”

Third case: Load = 100Ω

This is a typical case where the transmission line is connected to an unmatched load. Specifically, suppose we have a line with 50Ω impedance connected to a 100Ω load.



We proceed as in the previous cases.

$$V_0^+ = V_G \frac{Z_0}{Z_0 + R_S} = 5 \frac{50}{50 + 10} = 4.16 \text{ V}$$

Let's calculate the reflection coefficients at the source (Generator) and the load (Load).

$$\Gamma_G = \frac{Z_S - Z_0}{Z_S + Z_0} = \frac{10 - 50}{10 + 50} = -\frac{4}{6} = -\frac{2}{3}$$

$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{100 - 50}{100 + 50} = \frac{1}{3}$$

Let's recall the fundamental relationships:

$$\begin{aligned} V_n^- &= \Gamma_L V_n^+ \\ V_{n+1}^- &= \Gamma_G V_n^- = \Gamma_G \Gamma_L V_n^+ \end{aligned}$$

Let's calculate:

$$\Gamma_G \Gamma_L = \left(-\frac{2}{3}\right) \cdot \left(\frac{1}{3}\right) = -\frac{2}{9} = -0.222$$

The sequence of V_n^+ decreases in amplitude by a factor of $-\frac{2}{9}$ with alternating signs.

The voltage across the load (when the n^{th} packet arrives) is:

$$V_{Load,n} = V_n^+ + V_n^- = V_n^+ + \Gamma_L V_n^+ = (1 + \Gamma_L) V_n^+$$

$$\text{With } (1 + \Gamma_L) = 1 + \frac{1}{3} = \frac{4}{3}$$

The current in the load is:

$$I_{Load,n} = \frac{V_n^+ - V_n^-}{Z_0} = \frac{(1 - \Gamma_L) V_n^+}{Z_0}$$

$$\text{With } (1 - \Gamma_L) = \left(1 - \frac{1}{3}\right) = \frac{2}{3}$$

Iteration 0 (arrival at $t = T$)

- $V_0^+ = 4.1667 \text{ V}$
- $V_0^- = \Gamma_L V_0^+ = \left(\frac{1}{3}\right) \cdot 4.1667 \text{ V} = 1.3889 \text{ V}$

- $V_{Load,0} = (1 + \Gamma_L) V_0^+ = \left(\frac{4}{3}\right) \cdot 6.1667 \text{ V} = 5.5556 \text{ V}$
- $I_{Load,0} = \frac{(1 - \Gamma_L) V_0^+}{Z_0} = \frac{\frac{2}{3} \cdot 4.1667}{50} = 0.05556 \text{ A}$

Iteration 1 (arrival at $t = 3T$)

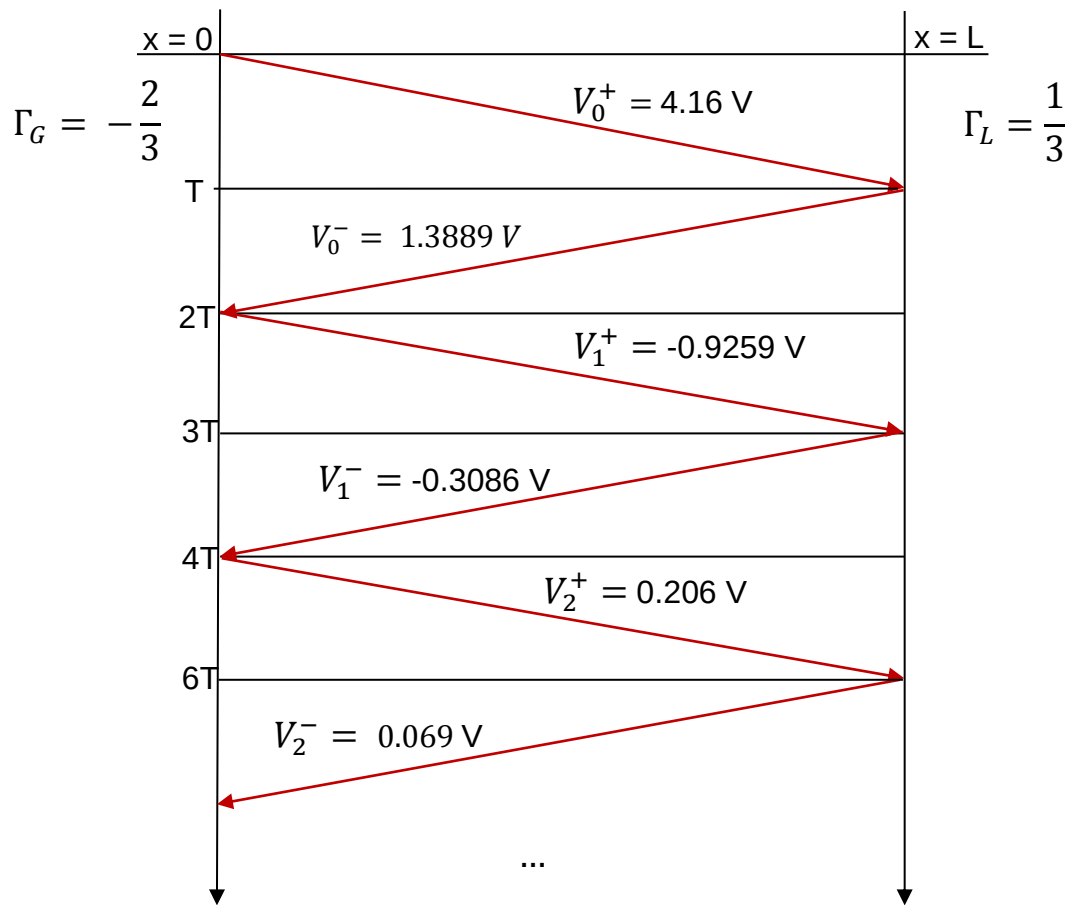
- $V_1^+ = \Gamma_G V_0^- = -\frac{2}{3} \cdot 1.3889 = -0.9259 \text{ V}$
- $V_1^- = \Gamma_L V_1^+ = \left(\frac{1}{3}\right) \cdot (-0.9259) = -0.3086 \text{ V}$
- $V_{Load,1} = (1 + \Gamma_L) V_1^+ = \left(\frac{4}{3}\right) \cdot (-0.9259) = -1.2346 \text{ V}$
- $I_{Load,1} = \frac{(1 - \Gamma_L) V_1^+}{Z_0} = \frac{\frac{2}{3} \cdot (-0.9259)}{50} = 0.01235 \text{ A}$

Iteration 2 (arrival at $t = 5T$)

- $V_2^+ = \Gamma_G V_1^- = -\frac{2}{3} \cdot (-0.3086) = 0.206 \text{ V}$
- $V_2^- = \left(\frac{1}{3}\right) \cdot 0.206 = 0.069 \text{ V}$
- $V_{Load,2} = (1 + \Gamma_L) V_2^+ = \left(\frac{4}{3}\right) \cdot 0.206 = 0.274 \text{ V}$
- $I_{Load,2} = \frac{(1 - \Gamma_L) V_2^+}{Z_0} = \frac{\frac{2}{3} \cdot (0.206)}{50} = 0.00274 \text{ A}$

Iteration 3 (arrival at $t = 7T$)

- $V_3^+ = \Gamma_G V_2^- = -\frac{2}{3} \cdot 0.069 = -0.046 \text{ V}$
- $V_3^- = \Gamma_L V_3^+ = \left(\frac{1}{3}\right) \cdot (-0.046) = -0.015 \text{ V}$
- $V_{Load,3} = (1 + \Gamma_L) V_3^+ = \left(\frac{4}{3}\right) \cdot (-0.046) = -0.061 \text{ V}$
- $I_{Load,3} = \frac{(1 - \Gamma_L) V_3^+}{Z_0} = \frac{\frac{2}{3} \cdot (-0.046)}{50} = -0.00061 \text{ A}$



Let's simulate the circuit.

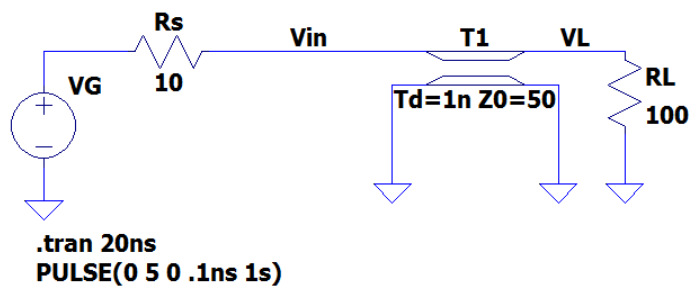


Figure 12: Circuit for simulation "3"

The following figure shows the behavior of the generator and load voltages. To better understand the behavior of these signals, the correspondences with the bounce diagram have been highlighted.

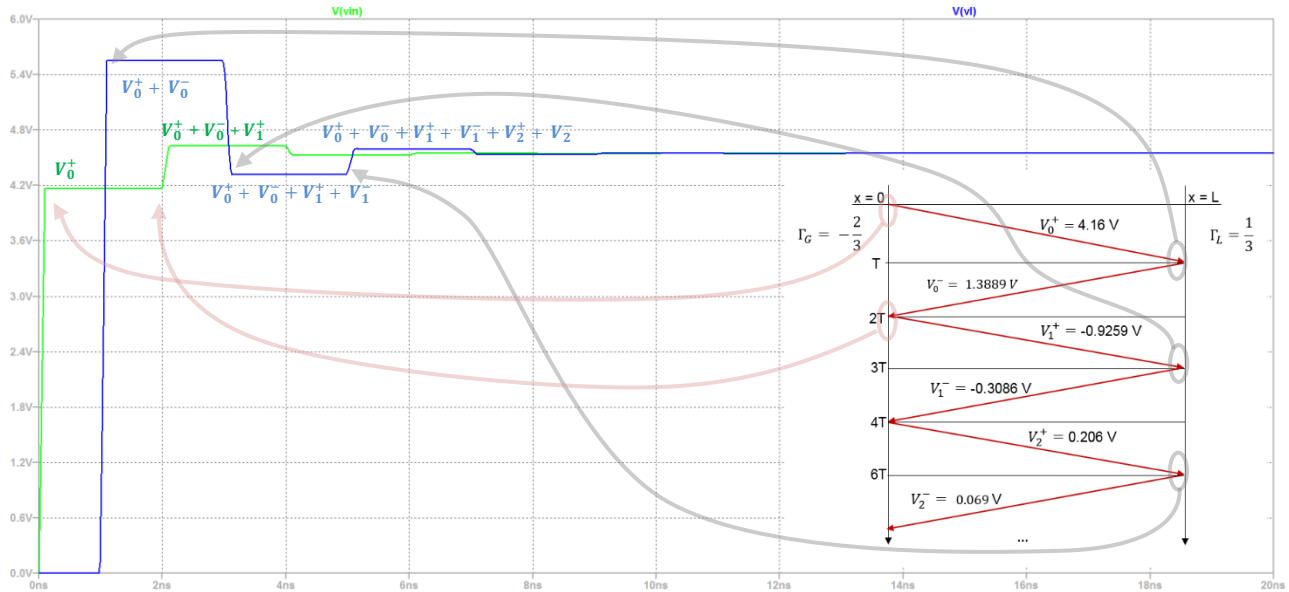
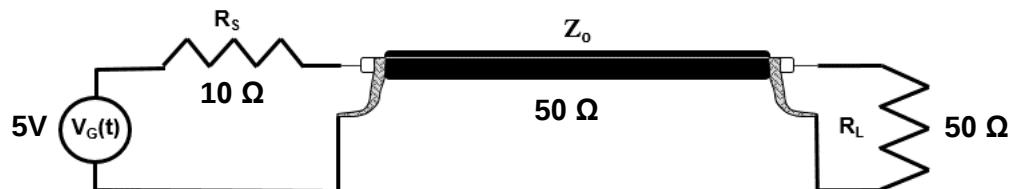


Figure 13: Result of the simulation of circuit "3"

Case Four: Load = 50Ω (impedance matching)

In this example, the load is perfectly matched to the transmission line.



If the load has a value equal to the characteristic impedance, there is no reflection and a perfect match occurs.

We proceed as in the previous cases.

$$V_0^+ = V_G \frac{Z_0}{Z_0 + R_s} = 5 \frac{50}{50 + 10} = 4.16 \text{ V}$$

Let's calculate the reflection coefficients at the source (Generator) and the load (Load).

$$\Gamma_G = \frac{Z_s - Z_0}{Z_s + Z_0} = \frac{10 - 50}{10 + 50} = -\frac{4}{6} = -\frac{2}{3}$$

$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{50 - 50}{50 + 50} = 0$$

Let's recall the fundamental relations:

$$\begin{aligned} V_n^- &= \Gamma_L V_n^+ \\ V_{n+1}^- &= \Gamma_G V_n^- = \Gamma_G \Gamma_L V_n^+ \end{aligned}$$

Let's calculate:

$$\Gamma_G \Gamma_L = \left(-\frac{2}{3}\right) \cdot 0 = 0$$

The sequence of V_n^- and V_{n+1}^+ is all zero for $n \geq 0$; only V_0^+ is nonzero.

The voltage at the load (when the n th packet arrives) is:

$$V_{Load,n} = V_n^+ + V_n^- = (1 + \Gamma_L) V_n^+ = (1 + 0) V_n^+ = V_n^+$$

The current in the load is:

$$I_{Load,n} = \frac{V_n^+ - V_n^-}{Z_0} = \frac{(1 - \Gamma_L) V_n^+}{Z_0} = \frac{(1 - 0) V_n^+}{Z_0} = \frac{V_n^+}{Z_0}$$

Iteration 0 (arrival at $t = T$)

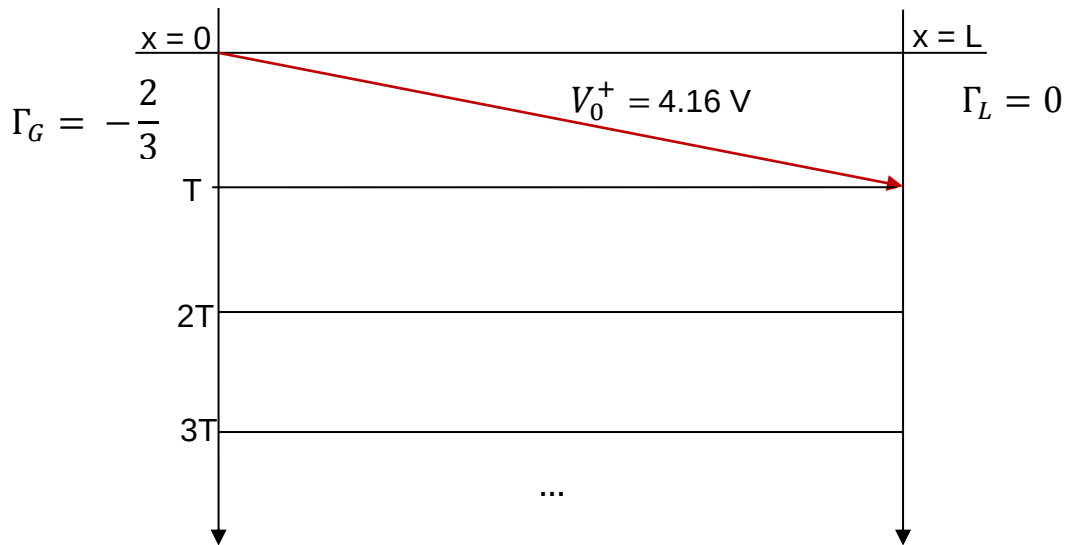
- $V_0^+ = 4.1667 \text{ V}$
- $V_0^- = V_0^+ \cdot \Gamma_L = 0 \text{ V}$ (remember that $\Gamma_L = 0$)
- $V_{Load,0} = (1 + \Gamma_L) V_0^+ = V_0^+ = 4.1667 \text{ V}$

- $I_{Load,0} = \frac{(1-\Gamma_L) V_0^+}{Z_0} = \frac{1 \cdot 4.1667}{50} = 0,0833 \text{ A}$

Iteration i = 1, 2, 3 (arrival at t = (2n+1)T)

- $V_i^+ = V_i^+ \cdot \Gamma_G = 0 \cdot \Gamma_G = 0V$
- $V_i^- = V_i^+ \cdot \Gamma_L = 0 \cdot \Gamma_L = 0V$

After the first iteration, there are no reflections because the load, perfectly matched to the transmission line, absorbs the entire V_0^+ , waveform. Therefore, after time T, the V_0^+ remains constant on the load.



Let's simulate the circuit.

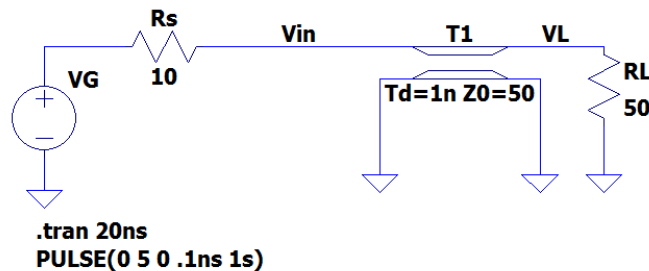


Figure 14: Circuit for simulation "4"

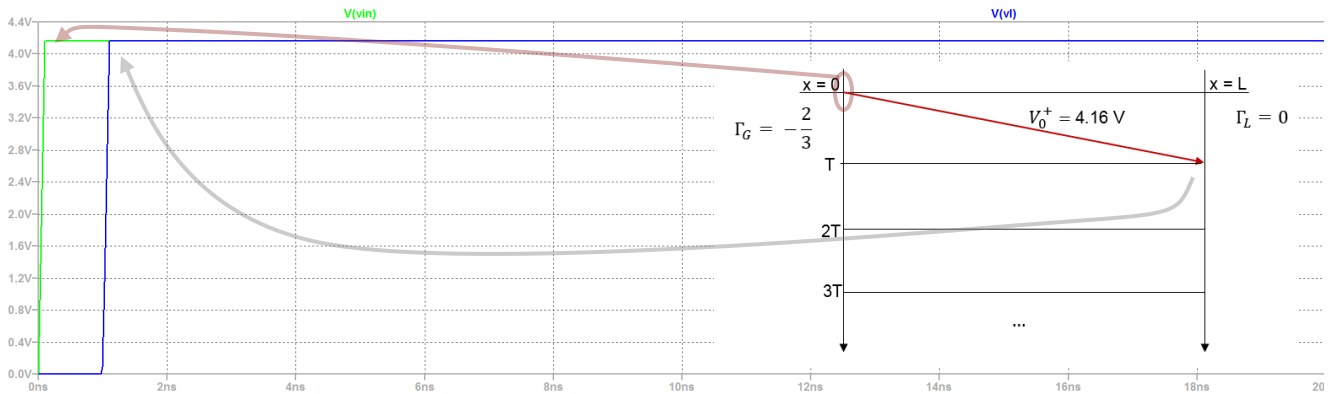


Figure 15: Simulation result of circuit "4"

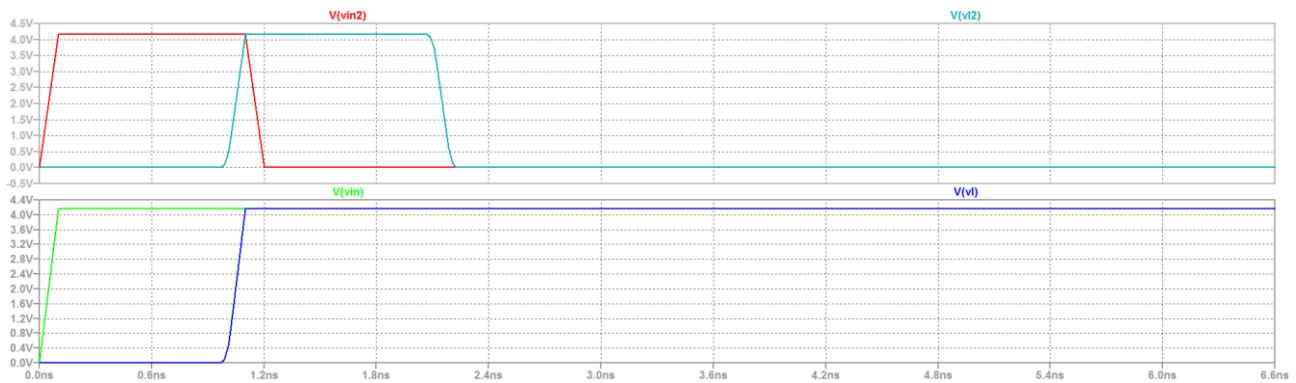
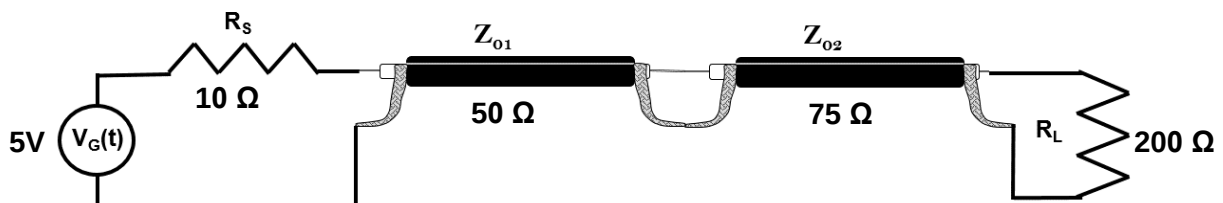


Figure 16: Response of circuit "4" to the step (bottom) and to the pulse (top)

Fifth case: complex system

Let's now examine a slightly more complex circuit, characterized by the presence of two transmission lines in series with different impedances. The load is not matched. We expect many "discontinuities" due to the signal passing through different impedances.



The methodology used to study the system remains unchanged.

First, let's define:

- Reflection coefficient of generator $\Gamma_G = \frac{R_S - Z_1}{R_S + Z_1} = -\frac{2}{3} = -0.667$
- Reflection coefficient of junction (1-2) $\Gamma_{1 \rightarrow 2} = \frac{Z_2 - Z_1}{Z_2 + Z_1} = \frac{25}{125} = 0.2$
- Reflection coefficient of junction (2-1) $\Gamma_{2 \rightarrow 1} = \frac{Z_1 - Z_2}{Z_1 + Z_2} = -0.2$
- Transmission coefficient $T_{1 \rightarrow 2} = 1 - \frac{Z_2 - Z_1}{Z_2 + Z_1} = \frac{2 Z_2}{Z_2 + Z_1} = \frac{150}{125} = 1.2$
- Transmission coefficient $T_{2 \rightarrow 1} = 1 - \frac{Z_1 - Z_2}{Z_2 + Z_1} = \frac{2 Z_1}{Z_2 + Z_1} = 0.8$
- Load reflection coefficient $\Gamma_L = \frac{Z_L - Z_2}{Z_L + Z_2} = \frac{125}{275} = 0.45$

Note that $T_{1 \rightarrow 2} > 1$. Why is a voltage greater than the incident voltage transmitted? This is consistent with energy conservation; just consider the impedances!

Let's now do some analysis on propagation times.

We have already seen that the propagation velocity $v_p = \frac{c}{\sqrt{\epsilon_r}} = \text{FV} \cdot c$, where $c = 3 \cdot 10^8 \frac{m}{s}$ and FV (velocity factor) depends on the transmission line (cable).

Suppose we have two coaxial cables with $\text{VF} = 0.66$, for example RG174 (50 Ω) and RG-59 (75 Ω); the first is 30 cm long and the second 90 cm. Let's calculate the wave propagation times in the two cables.

$$TD = \frac{L}{c \cdot \text{VF}} \rightarrow TD_1 = \frac{0.3 \text{ m}}{3 \cdot 10^8 \frac{m}{s} \cdot 0.66} \simeq 1.5 \text{ ns}, \quad TD_2 = \frac{0.9 \text{ m}}{3 \cdot 10^8 \frac{m}{s} \cdot 0.66} \simeq 4.5 \text{ ns}$$

Let's start calculating the reflection diagram, taking into account that:

$$TD_1 = 1.5 \text{ ns} \text{ e } TD_2 = 4.5 \text{ ns}$$

The first wave is always

$$V_{1,0}^+ = V_G \frac{Z_1}{R_S + Z_1} = 5 \frac{50}{10 + 50} = 4.1667 \text{ V}$$

This wave starts at $t=0$ and reaches the junction ($50\Omega \rightarrow 75\Omega$) at $t=TD_1=1.5 \text{ ns}$.

Iteration 0: $t = 1.5 \text{ ns}$

Incident wave on junction $V_{1,0}^+ = 4.167 \text{ V}$

- Reflection to the left

$$V_{1,0}^- = \Gamma_{1 \rightarrow 2} V_{1,0}^+ = 0.2 \cdot 4.167 = 0.833 \text{ V}$$

This wave returns to the source at $1.5 + 1.5 = 3 \text{ ns}$

- Transmission on line 2 to the load

$$V_{2,0}^+ = T_{1 \rightarrow 2} V_{1,0}^+ = 1.2 \cdot 4.167 = 5.0 \text{ V}$$

This wave reaches the load after a further $TD_2 = 4.5 \text{ ns}$, so, at:

$$1.5 + 4.5 = 6.0 \text{ ns}.$$

Iteration 1: $t = 3.0 \text{ ns}$

- The reflection $V_{1,0}^-$ reaches the generator and is reflected with $\Gamma_G = -\frac{2}{3}$.

$$V_{1,1}^+ = \Gamma_G V_{1,0}^- = -\frac{2}{3} \cdot 0.833 = -0.556 \text{ V}$$

This new forward wave starts at $t=3.0 \text{ ns}$ and arrives at the junction at $t=4.5 \text{ ns}$.

Iteration 2: $t = 4.5 \text{ ns}$

$V_{1,1}^+$ arrives at the junction.

- Reflection (negative sign) on line 1:

$$V_{1,1}^- = \Gamma_{1 \rightarrow 2} V_{1,1}^+ = 0.2 \cdot (-0.556) = -0.111 \text{ V}$$

It will return to the source and arrive at $4.5+1.5=6.0 \text{ ns}$.

- Transmission to the load on line 2:

$$V_{2,1}^+ = T_{1 \rightarrow 2} V_{1,1}^+ = 1.2 \cdot (-0.556) = -0.667 \text{ V}$$

which arrives at the load at $4.5+4.5=9.0 \text{ ns}$.

Iteration 3: t = 6.0 ns

Arrival of the first wave at the load and arrival of $V_{1,1}^-$ at the source.

- $V_{2,0}^+ = 5.0V$ arrives at the load.

Recall that $\Gamma_L = 0.45$

The wave reflected by the load is $V_{2,0}^- = \Gamma_L V_{2,0}^+ = 0.45 \cdot 5 = 2.27V$

The instantaneous voltage jump at the load (sum of incident + reflected) is:

$$\Delta V_{LOAD}(6ns) = (1 + \Gamma_L)V_{2,0}^+ = 1.45 \cdot 5 = 7.27V$$

Note that this is an overshoot with respect to V_G and is possible because the transmission to the junction has amplified the voltage.

- The reflected $V_{2,0}^-$ returns to the junction and reaches it at $6.0 + 4.5 = 10.5$ ns.
- At the same instant (t=6ns), the reflected wave $V_{1,1}^- = -0.11V$ arrives at the source. The generator reflects it again according to Γ_G generating:

$$V_{1,2}^+ = \Gamma_G V_{1,1}^- = -\frac{2}{3} \cdot 0.11 = 0.007V$$

This new wave arrives at the junction at $6.0+1.5=7.5$ ns.

Iteration 4: t = 7.5 ns

The junction receives $V_{1,2}^+$

- Reflection to the left: $V_{1,2}^- = 0.2 \cdot 0.07 = 0.015V$ which will return to the generator at 9.0 ns.
- Transmission to the load $V_{2,2}^+ = 1.2 \cdot 0.07 = 0.089V$ which will reach the load at $7.5 + 4.5 = 12.0$ ns.

Iteration 5: t = 9.0 ns

Arrival of $V_{2,1}^+$ at the load

- The voltage step at the load is: $\Delta V_{LOAD}(9ns) = (1 + \Gamma_L)V_{2,1}^+ = 0.97V$
The instantaneous value at the load, therefore, changes from 7.27V a 7.27-0.97 = 6.3V
- The generator receives $V_{1,2}^-$ and reflects it, creating

$$V_{1,3}^+ = \Gamma_G V_{1,2}^- = -\frac{2}{3} 0.015 = 0.0099V$$

Iteration 6: t = 10.5 ns

- The junction receives the reflected $V_{2,0}^-$ returned by the load (at the same time, a small wave also arrives at the node from the left).
- Here, we need to use superposition: at the junction, there are two simultaneous incident waves:
 - From the right $a_2 = V_{2,0}^- = 2,27V$
 - From the left $a_1 = V_{1,3}^+ = 0,0099 V$

So, let's define:

- $b_1 = \Gamma_{1 \rightarrow 2} a_1 + T_{2 \rightarrow 1} a_2 = 1,81 V$ (goes to the source and arrives at 10.5+1.5=12.0 ns)
- $b_2 = T_{1 \rightarrow 2} a_1 + \Gamma_{2 \rightarrow 1} a_2 = -0.466V$ (goes to the load and arrives at 10.5+4.5=15.0 ns)

These are the first iterations, shown to illustrate how the circuit works. Every time a wave reaches one of the three points (source/junction/load), a transmitted and a reflected contribution is generated; the contributions add linearly over time at the node we are observing.

The process repeats until all reflections are eliminated.

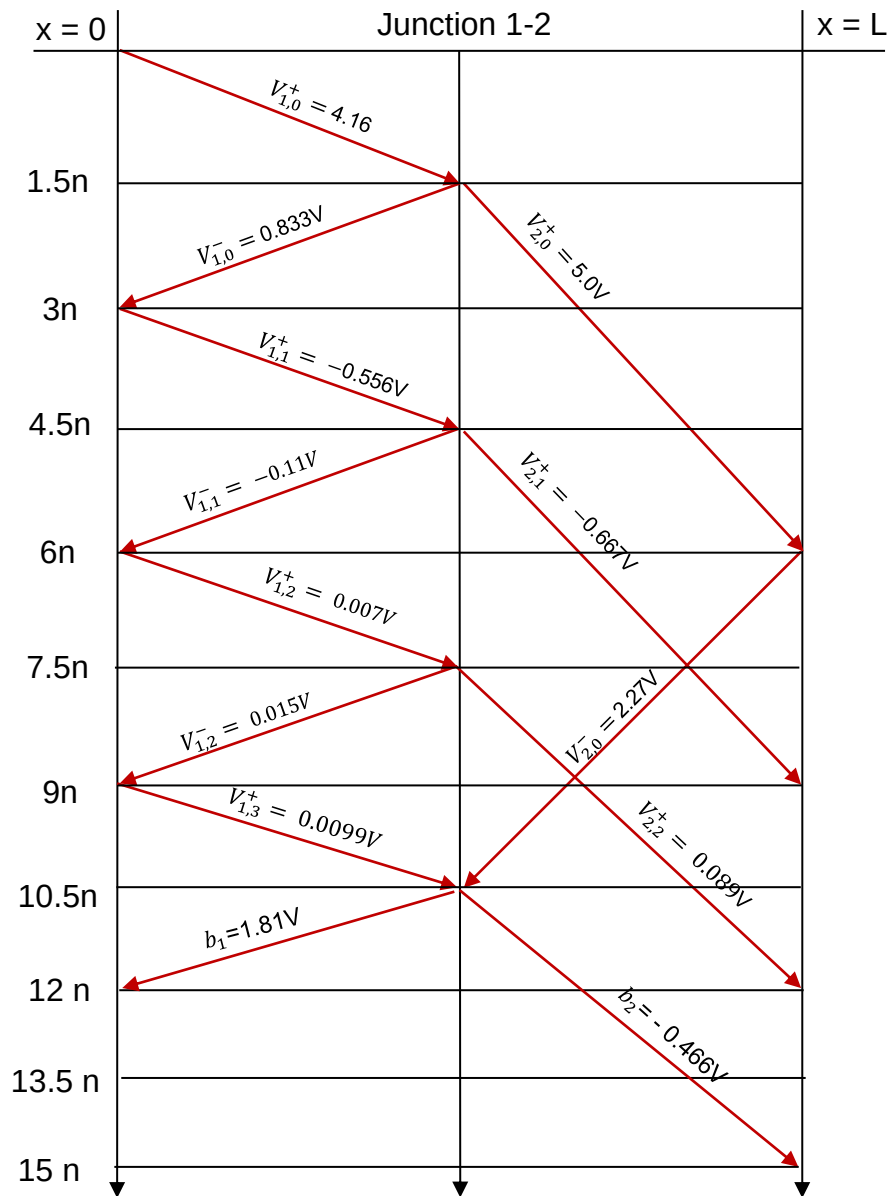
Where does the system converge?

After a sufficiently long time (many bounces), the line behaves like a DC connection to the load Z_L : the steady state DC is a simple divider between R_S and Z_L :

$$V_{Load,\infty} = V_G \frac{Z_L}{R_S + Z_L} = 5 \frac{200}{10 + 200} = 4.76V$$

$$I_{Load,\infty} = \frac{V_G}{R_S + Z_L} = 0.024A$$

The calculated iterations are shown in the following graph.



We can now simulate the circuit.

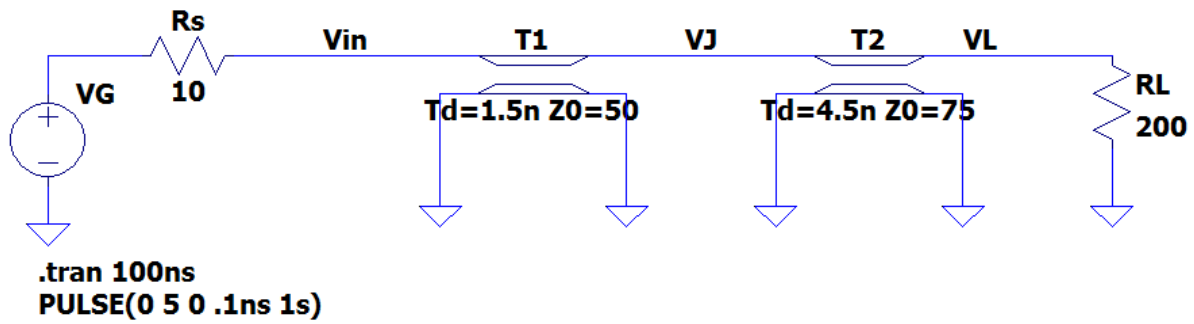
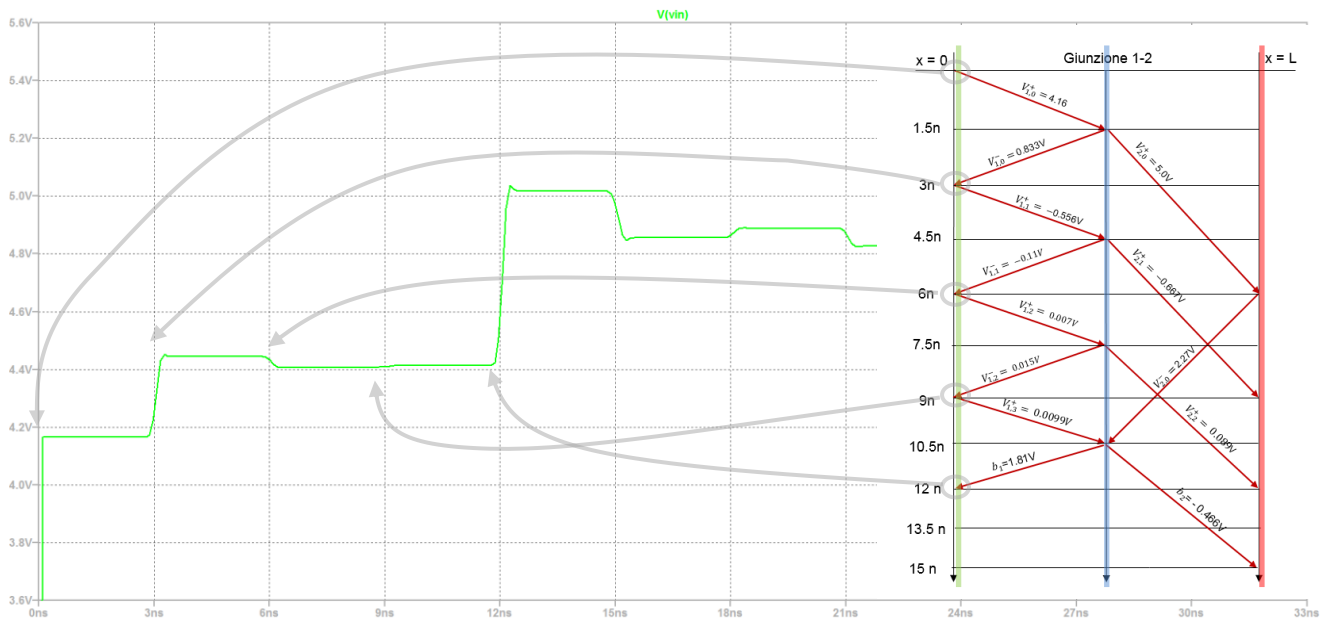


Figure 17: Simulation circuit for circuit "5"

The following figure shows the time graph of the load voltage. The events reported in the bounce graph have been highlighted to explain the different voltage levels.



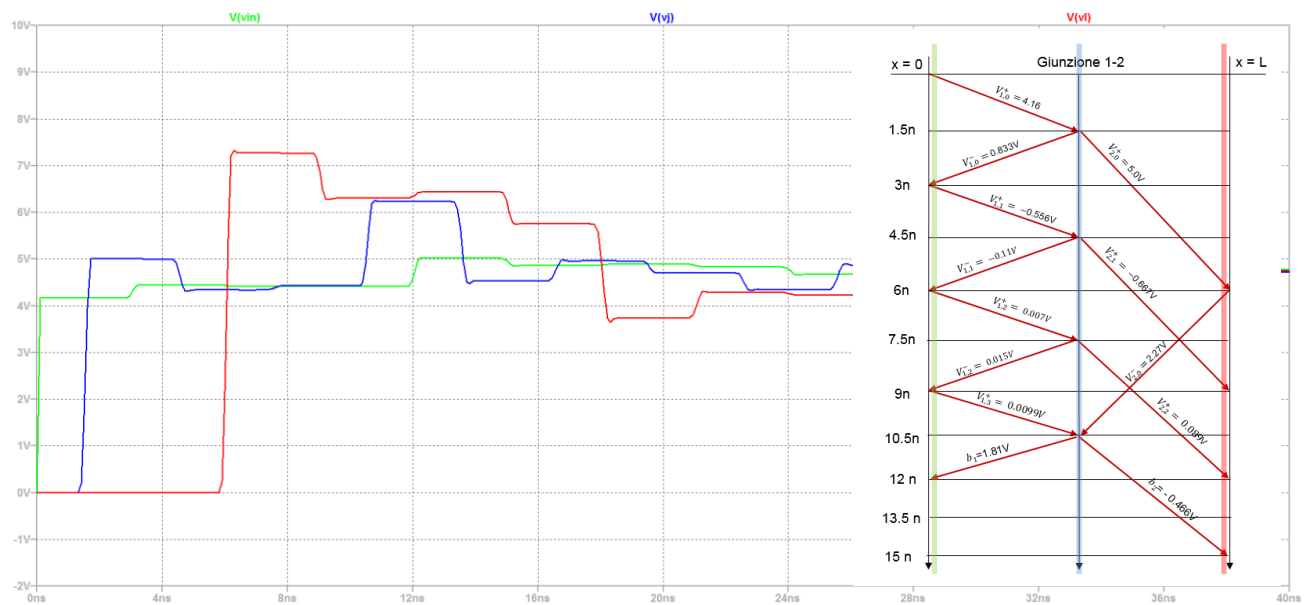


Figure 19: Simulation results for circuit "5", with voltages at the source, junction, and load

4. Conclusions

Time-domain reflectometry, which we've explored in this tutorial from a theoretical, practical, and applied perspective, represents one of the most powerful and fascinating techniques for studying transmission lines.

Every time an electrical signal travels along a non-ideal path, be it a coaxial cable, a printed circuit board, or a twisted pair, it is inevitably subject to reflections, attenuation, mismatches, and dispersion. Under ideal conditions, a signal sent from the source to the load should never "return": all the power should be absorbed by the load, with no reflection back to the source. But the real world is far from perfect. Loads aren't always matched, lines can exhibit discontinuities, imperfect connections, spurious branches, and impedance variations due to manufacturing technology.

This is why reflectometry becomes a tool that allows the technician to study the line and observe what happens inside it, without having to physically disassemble it. By sending a known pulse or step and analyzing the returned reflections, we can reconstruct a spatial map of the discontinuities, almost as if we had a miniature radar scanning the inside of the cable or track.

During our journey, we've seen how TDR is based on fundamental theoretical concepts in electrical engineering: the theory of distributed-parameter transmission lines, the definition of characteristic impedance, the reflection coefficient, the distinction between geometric and electrical length, and spatial resolution. All of this constitutes the "language" with which we can describe the phenomena we observe.

Theory, however, remains sterile unless tested with models and simulations. This is why we introduced the bounce diagram tool, which allows us to intuitively visualize how waves propagate along the line, with continuous reflections and transmissions. This method, although simplified, provides an immediate understanding of what would otherwise remain confined to complex equations.

Subsequently, using LTspice, we saw how mathematical models can be made concrete, giving rise to simulations that faithfully reproduce the time evolution of voltages and currents.